

Exact Randomized Designs for Two-Block Interference Experiments

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Abstract

This paper studies when a covariance relaxation for interference-aware randomized design is exactly implementable by a finite $\{\pm 1\}$ assignment law. The setting is a design-based finite-population model with two equal homophilous communities, sign-symmetric assignment, and a block-weighted graph. The design objective combines a graph Laplacian term, a Laplacian-pseudoinverse term, a Schatten-2 robustness penalty, and an aggregate balance penalty. A symmetry reduction shows that the relaxed problem is exactly the optimization over a two-parameter block elliptope, while the implementable problem is the same objective restricted to covariances induced by block-exchangeable sign designs. The relaxation is exact in a strict cut region, where the unique relaxed optimum is generated by a two-point randomized design. Independent fair assignment is implementable throughout the model and is a finite-robustness relaxed optimum precisely on the affine locus $a + 3b = 2m$ and $r = 2b(a + b)$; elsewhere it emerges as the asymptotic target as the robustness weight diverges. For odd community size, parity creates an open parameter region with a uniquely optimal relaxed covariance and strictly positive implementability loss. Finally, an exact finite active-set formula computes the loss for all admissible m, a, b, r, κ , with zero loss for even community size.

1 Introduction

Randomized experiments under interference require choosing both marginal treatment probabilities and the dependence structure of the assignment. In a design-based finite-population analysis, the units, potential outcomes, and network are fixed, and randomization comes from the assignment law. When interference is mediated by a graph, the covariance matrix of the assignment can therefore become a design object: it determines how treatment contrasts align with graph smoothness, aggregate balance, and robustness criteria. This perspective is closely related to the randomization-based tradition in causal inference [Horvitz and Thompson, 1952, Rubin, 1974], to design and inference under interference [Hudgens and Halloran, 2008, Aronow and Samii, 2013, Sävje et al., 2017, Leung, 2019, Li and Wager, 2020], and to recent covariance-based design formulations for network experiments [Thiyageswaran et al., 2026].

A covariance relaxation is attractive because it replaces a finite randomized assignment problem by a convex matrix problem. Its feasible set can be strictly larger than the covariances generated by actual $\{\pm 1\}$ assignment laws, producing the familiar gap between semidefinite or elliptope relaxations and discrete correlation structures [Goemans and Williamson, 1995, Deza and Laurent, 1997, Boyd and Vandenberghe, 2004]. In experimental design, this gap

has an operational meaning: implementation requires a covariance prescription induced by a randomization mechanism.

This paper isolates that issue in a finite two-community model. There are $n = 2m$ fixed units, split into two equal communities. The graph has within-community edge weight a/m and across-community edge weight b/m , with $a > b > 0$. Assignments are sign vectors $Z \in \{-1, 1\}^n$, and assignment laws are required to be sign-symmetric. The objective $F_{r,\kappa}$ in [definition 6](#) combines four terms: graph Laplacian exposure, a Laplacian-pseudoinverse component weighted by r , a Schatten-2 robustness penalty weighted by κ , and an aggregate balance penalty. The object of interest is the implementability loss $\Delta_m^\pm(r, \kappa)$, the difference between the best implementable block-exchangeable sign design and the relaxed block-elliptope benchmark. [Lemma 1](#) establishes equality with the infimum over the broader sign-symmetric class $\mathcal{P}_m^{\text{bal}}$.

The first step is a symmetry reduction. Under [assumption 1](#), [lemma 1](#) shows that averaging over graph symmetries weakly decreases the objective for both relaxed covariance matrices and sign-symmetric assignment laws. Hence the relaxed infimum is exactly the infimum over $\mathcal{E}_m^{\text{blk}}$, and the implementable infimum is exactly the infimum over $\mathcal{P}_m^{\text{sym}}$. This reduction makes the implementability question finite-dimensional.

The paper then gives two exactness results. Under [assumption 1](#), [theorem 1](#) shows that, for $r, \kappa \geq 0$ satisfying the displayed strict cut-region inequality, the cut covariance X_{cut} is the unique relaxed minimizer. It is also generated by the symmetric two-point cut design, so the implementability loss is zero throughout that region. This exact finite-sample result supplies a transparent sufficient region for literal implementation of the relaxed optimum.

The second exactness result concerns iid assignment. Under [assumption 1](#) and $r \geq 0$, [theorem 2](#) shows that I_n is a finite-robustness relaxed minimizer precisely when $a + 3b = 2m$ and $r = 2b(a + b)$. On that affine-balanced locus, every positive κ makes I_n the unique relaxed minimizer. Elsewhere, relaxed minimizers converge entrywise to I_n as $\kappa \rightarrow \infty$. Thus iid assignment is always implementable in this symmetric class, with exact finite- κ optimality characterized by the affine-balanced locus.

The main positive-loss result is arithmetic. When m is odd, each realized block sum has odd parity and therefore absolute value at least one. Under the homophily, low-scale, and odd-community conditions, [theorem 3](#) identifies a nonempty robustness range and an open interval of pseudoinverse weights in which the spread covariance X_{spread} is the unique relaxed minimizer, lies outside $\mathcal{C}_m^\pm$, and yields $\Delta_m^\pm(r, \kappa) > 0$. The same theorem also shows that every covariance in the block elliptope is implementable when m is even.

For computation outside that local positive-gap window, [theorem 4](#) gives an exact finite formula. Under [assumption 1](#), for every $r, \kappa \geq 0$, the loss equals a sharp reduced quantity $\rho_*(m, a, b, r, \kappa)$. The relaxed problem becomes a weighted simplex problem in three spectral coordinates, and implementability adds only the parity truncation $d_m \leq y + z_{\text{sp}}$, where $d_m = 0$ for even m and $d_m = 2/m$ for odd m . With positive robustness, the relaxed minimizer is unique and is obtained by an active-set calculation; if it violates the parity truncation, the implementable value is obtained on a one-dimensional boundary segment. With $\kappa = 0$, the theorem gives the corresponding exposed-face condition. The formal statements and proofs are machine-checked in Lean 4 to the extent described in the verification appendix.

The contribution is an exact finite-sample study of the covariance criterion $F_{r,\kappa}$ in a homophilous two-block design. It characterizes literal implementability and positive loss and computes the exact objective-value price of returning from the relaxation to a feasible randomized assignment. Every statement in the formal layer is machine-checked in Lean 4 (Appendix B).

2 Setup and assumptions

We work in a design-based finite-population assignment model: the units are fixed, and the only random object is the treatment assignment. There are $n = 2m$ units partitioned into two equal communities. An assignment is represented by the sign vector $Z \in \{-1, 1\}^n$, with a realized assignment denoted by z , and an assignment law P is a probability mass function on this finite assignment space. This formulation follows the randomization-based tradition in which the assignment mechanism is the source of sampling variation [Horvitz and Thompson, 1952, Rubin, 1974], while the network structure enters through the design criterion rather than through a superpopulation model.

The graph used by the criterion is a two-community weighted graph. The analysis rests first on the following homophily and scale conventions.

Assumption 1. [ass:two-block-homophily] (Two-Block Homophily) *The block size and edge intensities satisfy*

$$m \geq 2 \quad \text{and} \quad a > b > 0.$$

The constants a and b encode stronger interaction intensity within a community than across communities. This homophily restriction is specific to this analysis: it gives the two-block model a nondegenerate within-versus-between contrast while keeping both edge intensities positive.

Definition 1. [def:two-block-graph] (Two-Block Graph Weights W_{ij}) *The vertex set is the disjoint union $A_m \cup B_m$, where*

$$|A_m| = |B_m| = m.$$

The graph weights W_{ij} are given as follows: if $i \neq j$ and i and j lie in the same block, then

$$W_{ij} = \frac{a}{m};$$

if i and j lie in opposite blocks, then

$$W_{ij} = \frac{b}{m};$$

and for every vertex i ,

$$W_{ii} = 0.$$

Thus the communities A_m and B_m have equal size, and all heterogeneity in exposure strength is summarized by whether a pair lies within or across communities. Let L_m denote the graph Laplacian associated with these weights, with diagonal entries equal to weighted degrees and off-diagonal entries $-W_{ij}$. We write $L_m^\dagger$ for its Moore–Penrose pseudoinverse. Network-interference applications often use such weighted graphs to encode exposure or spillover structure [Hudgens and Halloran, 2008, Aronow and Samii, 2013, Sävje et al., 2017, Leung, 2019, Li and Wager, 2020]; here the two-block graph isolates the exact finite-sample geometry of a homophilous special case.

The sign representation is invariant to a global relabeling of treatment and control. We impose that invariance directly on the assignment law.

Assumption 2. [ass:balanced-sign-design] (Balanced Sign Symmetry) *The assignment law is sign-symmetric: for every $z \in \{-1, 1\}^n$,*

$$P(Z = z) = P(Z = -z).$$

This symmetry condition is specific to this analysis. It expresses balance between the two treatment labels at the law level, yields zero one-point margins, and permits many forms of dependence and varying treated counts across realizations.

Definition 2. [def:balanced-design-class] (Balanced sign class $\mathcal{P}_m^{\text{bal}}$) Under [assumption 2](#), the balanced sign class is

$$\mathcal{P}_m^{\text{bal}} = \{P \text{ on } \{-1, 1\}^n : P(Z = z) = P(Z = -z) \text{ for every } z \in \{-1, 1\}^n\}.$$

The class $\mathcal{P}_m^{\text{bal}}$ contains all sign-symmetric laws. For the two-block graph, the relevant symmetric subclass also treats units within a block exchangeably and treats the two equal blocks symmetrically.

Definition 3. [def:block-exchangeable-design-class] (Block-exchangeable class $\mathcal{P}_m^{\text{sym}}$) The block-exchangeable class is the set of all assignment laws

$$P \in \mathcal{P}_m^{\text{bal}}$$

that are invariant under permutations within A_m , invariant under permutations within B_m , and invariant under swapping the two equal communities.

For optimization of $F_{r,\kappa}$ over the sign-symmetric class $\mathcal{P}_m^{\text{bal}}$, the block-exchangeable class $\mathcal{P}_m^{\text{sym}}$ preserves the optimum: averaging over the two-block automorphism group weakly decreases this invariant objective. This reduction makes the maintained implementability question finite-dimensional.

For any assignment law, write $X(P)$ for the second-moment matrix,

$$X(P) = \mathbb{E}_P[ZZ^\top].$$

This is a unit-diagonal second-moment (correlation) matrix; under the sign symmetry of [assumption 2](#) the one-point margins vanish, so for implementable laws it coincides with the covariance of Z . We refer to X as a second-moment or correlation matrix throughout, reserving “covariance” for the sign-symmetric case; the relaxed matrices below are unit-diagonal correlation matrices that need not arise from any law. The covariance relaxation replaces such implementable second moments by block-symmetric correlation matrices. For scalars u and v , $X(u, v)$ denotes the matrix with diagonal entries equal to one, common off-diagonal covariance u within each block, and common covariance v across the two blocks.

Definition 4. [def:block-elliptope] (Block Elliptope $\mathcal{E}_m^{\text{blk}}$) Under [assumption 1](#), the block elliptope is

$$\mathcal{E}_m^{\text{blk}} = \{X(u, v) : 1 - u \geq 0, 1 + (m - 1)u - mv \geq 0, 1 + (m - 1)u + mv \geq 0\}.$$

The relaxed covariance set $\mathcal{E}_m^{\text{blk}}$ is the block-symmetric slice of the elliptope. The three displayed inequalities are the positive-semidefiniteness restrictions in the invariant directions induced by the two equal blocks. This convex relaxation is the object optimized before imposing that the covariance arise from an actual $\{\pm 1\}$ assignment law [[Boyd and Vandenberghe, 2004](#), [Thiyageswaran et al., 2026](#)].

Definition 5. [def:implementable-covariance-class] (Implementable covariance class $\mathcal{C}_m^\pm$) The implementable covariance class is

$$\mathcal{C}_m^\pm = \{X(P) : P \in \mathcal{P}_m^{\text{sym}}\}.$$

The set $\mathcal{C}_m^\pm$ records the discrete feasibility constraint that remains after symmetry reduction: a relaxed covariance matrix is useful for exact randomized design only when it is generated by some block-exchangeable sign law.

The objective combines graph smoothness, a pseudoinverse term, robustness, and a balance penalty. The scalar r weights the pseudoinverse term, κ weights the Schatten–2 norm penalty, and J_n denotes the $n \times n$ all-ones matrix.

Definition 6. [def:design-objective] (Design objective $F_{r,\kappa}$) *For $r, \kappa \geq 0$, the design objective evaluated at a covariance matrix X is*

$$F_{r,\kappa}(X) = \text{tr}(L_m X) + r \text{tr}(L_m^\dagger X) + \kappa \|X\|_{S_2} + \text{tr}(J_n X).$$

The first term is the expected weighted disagreement of the sign assignment across graph edges: because $\text{tr}(L_m X) = \sum_{i < j} W_{ij} \mathbb{E}[(Z_i - Z_j)^2] = 2 \sum_{i < j} W_{ij} (1 - X_{ij})$ for a unit-diagonal second moment, minimizing it penalizes sign dispersion across high-weight edges and rewards positive treatment-sign alignment there. The second term uses the Laplacian pseudoinverse to weight low-frequency graph directions, the third is a Frobenius-norm robustness penalty, and the final term $\text{tr}(J_n X) = \mathbb{E}[(\sum_i Z_i)^2]$ penalizes imbalance in the aggregate sign vector. We treat $F_{r,\kappa}$ as a covariance design criterion in the spirit of the network-experiment SDP formulation of [Thiyageswaran et al. \[2026\]](#). The paper compares this criterion over the relaxed block ellipsope and implementable sign designs and characterizes exactly when the optima coincide. The implementability loss measures their objective-value gap.

Definition 7. [def:implementability-gap] (Implementability loss $\Delta_m^\pm(r, \kappa)$) *The implementability loss is*

$$\Delta_m^\pm(r, \kappa) = \inf_{P \in \mathcal{P}_m^{\text{sym}}} F_{r,\kappa}(X(P)) - \inf_{X \in \mathcal{E}_m^{\text{blk}}} F_{r,\kappa}(X).$$

The quantity $\Delta_m^\pm(r, \kappa)$ is therefore the exact price of requiring a covariance matrix to be induced by a finite randomized assignment law rather than merely satisfying the block ellipsope constraints.

A useful handle on implementability is provided by the two block sums. For a given law P , these sums retain the information relevant to the block-symmetric second moments.

Definition 8. [def:block-sum-handle] (Block-sum law $\mathcal{L}_P(S_A, S_B)$) *For an assignment law P , the block-sum law is the law under P of*

$$(S_A, S_B) = \left(\sum_{i \in A_m} Z_i, \sum_{i \in B_m} Z_i \right).$$

The block-sum law $\mathcal{L}_P(S_A, S_B)$ will be used to express the discrete restrictions that separate $\mathcal{C}_m^\pm$ from $\mathcal{E}_m^{\text{blk}}$.

Two additional restrictions are stated here because later exact-loss results invoke them explicitly.

Assumption 3. [ass:low-scale-two-block] (Low-Scale Two-Block Regime) *The two-block edge intensities satisfy*

$$a + 3b < 2m.$$

The low-scale regime is specific to this analysis. It is a finite-sample scale restriction comparing graph intensity with block size; for fixed a and b , it is automatically satisfied once m is large enough.

Assumption 4. [ass:odd-community-size] (Odd Community Size) *The block size m is odd.*

The odd-block condition is also specific to this analysis. It isolates the parity case in which every realized block sum has absolute value at least one, which is exactly where the later implementability loss result has content.

The following reduction justifies working with the block elliptope and the block-exchangeable assignment class while preserving the relevant optima.

Lemma 1. [prop:symmetry-reduction] (Symmetry Reduction Principle) *Assume [assumption 1](#). Let $r, \kappa \in [0, \infty)$. Then:*

- **(Relaxed matrices.)** *For every positive semidefinite matrix $X \in \mathbb{R}^{n \times n}$ with $X_{ii} = 1$ for every i , let u and v be the averages of the off-diagonal within-community and across-community entries of X , respectively. The block matrix $X(u, v)$ of [definition 4](#) has feasible block-elliptope parameters, and*

$$F_{r, \kappa}(X(u, v)) \leq F_{r, \kappa}(X),$$

where $F_{r, \kappa}$ is [definition 6](#).

- **(Assignment laws.)** *For every assignment law $P \in \mathcal{P}_m^{\text{bal}}$ in the sense of [definition 2](#), its orbit average over the full two-block automorphism group belongs to $\mathcal{P}_m^{\text{sym}}$ in the sense of [definition 3](#), has zero one-point margins,*

$$\mathbb{E}[Z_i] = 0 \quad \text{for every } i,$$

and satisfies

$$F_{r, \kappa}(X(P^{\text{orb}})) \leq F_{r, \kappa}(X(P)).$$

- **(Relaxed infimum.)**

$$\inf_{\substack{X \succeq 0 \\ X_{ii} = 1 \ \forall i}} F_{r, \kappa}(X) = \inf_{X \in \mathcal{E}_m^{\text{blk}}} F_{r, \kappa}(X).$$

- **(Implementable infimum.)**

$$\inf_{P \in \mathcal{P}_m^{\text{bal}}} F_{r, \kappa}(X(P)) = \inf_{P \in \mathcal{P}_m^{\text{sym}}} F_{r, \kappa}(X(P)).$$

The proof is deferred to [appendix C](#).

[Lemma 1](#) is the bridge between the original finite assignment problem and the two-parameter geometry used in the rest of the paper. It says that, for the objective in [definition 6](#), symmetrizing either a relaxed covariance matrix or an assignment law is weakly improving. Consequently, the relaxed benchmark is exactly the infimum over $\mathcal{E}_m^{\text{blk}}$, and the implementable benchmark is exactly the infimum over $\mathcal{P}_m^{\text{sym}}$.

3 Exactness at implementable designs

The symmetry reduction in [lemma 1](#) leaves a precise question: when does the relaxed optimum over $\mathcal{E}_m^{\text{blk}}$ coincide with a covariance matrix generated by an actual block-exchangeable sign design? This section records two exact cases. We use three block covariances: the cut covariance $X_{\text{cut}} = X(1, -1)$, the identity $I_n = X(0, 0)$ corresponding to independent fair Bernoulli signs, and the spread covariance $X_{\text{spread}} = X\left(-\frac{1}{m-1}, 0\right)$. The implementing laws are the two-point cut law P_{cut} and the independent fair-sign law P_{iid} . These designs represent different design instincts: one separates the homophilous communities, while the other spreads randomization independently across all units [[Efron, 1971](#), [Morgan and Rubin, 2012](#), [Harshaw et al., 2019](#)].

The cut design is optimal in a strict small- r region. There, the graph and balance terms favor separating the two blocks strongly enough to keep the relaxed optimum at the cut corner after including the pseudoinverse and robustness terms.

Theorem 1. [thm:cut-corner-exactness] (Cut-Corner Exactness) *Assume the two-block homophily condition in [assumption 1](#). Let $r, \kappa \in \mathbb{R}$ satisfy*

- (Nonnegative tradeoff.) $0 \leq r$.
- (Nonnegative robustness.) $0 \leq \kappa$.
- (Cut region.)

$$r < r_{\text{cut}}(m, a, b, \kappa) := \max\left\{0, \min\left(2b(a+b)\left(1 - \frac{\kappa}{a-b}\right), 2b(2m-2b-\kappa)\right)\right\}.$$

Then $X_{\text{cut}} \in \mathcal{E}_m^{\text{blk}}$, and for every $X \in \mathcal{E}_m^{\text{blk}}$ with $X \neq X_{\text{cut}}$,

$$F_{r,\kappa}(X_{\text{cut}}) < F_{r,\kappa}(X),$$

where $F_{r,\kappa}$ is the objective in [definition 6](#). Moreover,

$$\Delta_m^\pm(r, \kappa) = 0,$$

with $\Delta_m^\pm$ as in [definition 7](#). Finally, $X_{\text{cut}} \in \mathcal{C}_m^\pm$, and for every $X \in \mathcal{C}_m^\pm$ with $X \neq X_{\text{cut}}$,

$$F_{r,\kappa}(X_{\text{cut}}) < F_{r,\kappa}(X),$$

where $\mathcal{C}_m^\pm$ is the class in [definition 5](#); the cut design satisfies

$$P_{\text{cut}} \in \mathcal{P}_m^{\text{sym}} \quad \text{and} \quad X(P_{\text{cut}}) = X_{\text{cut}},$$

with $\mathcal{P}_m^{\text{sym}}$ as in [definition 3](#).

[theorem 1](#) is an exact attainability statement. The displayed strict inequality involving $r_{\text{cut}}(m, a, b, \kappa)$ gives a sufficient region in which the relaxed block-elliptope optimum is unique, implementable, and has zero implementability loss. The implementing law P_{cut} is the symmetric two-point law on the cut assignment and its global sign reversal. This result characterizes optimality for the stated covariance criterion $F_{r,\kappa}$.

The second exactness result concerns the robust corner I_n . Independent fair assignment is always implementable, but the theorem below shows that its relaxed optimality is much more restrictive: a finite positive robustness weight can make I_n optimal only on a particular affine relation among the graph scale and the pseudoinverse weight.

Theorem 2. [thm:robust-corner-exactness] (Robust Corner Exactness) *Let $m \in \mathbb{N}$ and $a, b, r \in \mathbb{R}$. Suppose that*

- *(Two-block homophily.) [assumption 1](#) holds for (m, a, b) .*
- *(Nonnegative pseudoinverse weight.) $r \geq 0$.*

For the objective $F_{r,\kappa}$ of [definition 6](#), the following statements hold.

First, there exists a finite nonnegative robustness weight κ such that $I_n \in \mathcal{E}_m^{\text{blk}}$ and

$$F_{r,\kappa}(I_n) \leq F_{r,\kappa}(X) \quad \text{for every } X \in \mathcal{E}_m^{\text{blk}},$$

if and only if

$$a + 3b = 2m \quad \text{and} \quad r = 2b(a + b).$$

Second, on this affine-balanced locus, every positive robustness weight makes I_n the unique relaxed minimizer: if

$$a + 3b = 2m \quad \text{and} \quad r = 2b(a + b),$$

then for every $\kappa > 0$,

$$I_n \in \mathcal{E}_m^{\text{blk}}$$

and, for every $X \in \mathcal{E}_m^{\text{blk}}$ with $X \neq I_n$,

$$F_{r,\kappa}(I_n) < F_{r,\kappa}(X).$$

Third, the iid design satisfies

$$P_{\text{iid}} \in \mathcal{P}_m^{\text{sym}}$$

for the block-exchangeable class of [definition 3](#), and

$$X(P_{\text{iid}}) = I_n.$$

Finally, off the affine-balanced locus, I_n is never a minimizer at any positive finite robustness weight, and relaxed minimizers converge entrywise to I_n as $\kappa \rightarrow \infty$. Precisely, if

$$\neg(a + 3b = 2m \text{ and } r = 2b(a + b)),$$

then for every $\kappa > 0$ it is not the case that

$$F_{r,\kappa}(I_n) \leq F_{r,\kappa}(X) \quad \text{for every } X \in \mathcal{E}_m^{\text{blk}}.$$

Moreover, for any family $(X_\kappa)_{\kappa \in \mathbb{R}}$ such that, for every $\kappa > 0$,

$$X_\kappa \in \mathcal{E}_m^{\text{blk}} \quad \text{and} \quad F_{r,\kappa}(X_\kappa) \leq F_{r,\kappa}(X) \quad \text{for every } X \in \mathcal{E}_m^{\text{blk}},$$

each matrix entry converges to the corresponding entry of I_n :

$$(X_\kappa)_{ij} \longrightarrow (I_n)_{ij} \quad \text{as } \kappa \rightarrow \infty$$

for every $i, j \in \{1, \dots, n\}$.

[theorem 2](#) separates implementability from optimality. The iid law P_{iid} belongs to the block-exchangeable design class and attains I_n , but this implementable covariance is a relaxed minimizer at finite robustness only when $a + 3b = 2m$ and $r = 2b(a + b)$. Off that affine-balanced locus, relaxed minimizers converge entrywise to I_n as $\kappa \rightarrow \infty$, yet no positive finite value of κ makes I_n itself optimal. Thus robustness can make iid assignment an asymptotic limiting target without making it an exact finite- κ solution.

Together, [theorem 1](#) and [theorem 2](#) identify two implementable covariances whose roles are sharply different. The cut covariance is exactly optimal throughout the stated strict cut region, yielding zero implementability loss there. The iid covariance is implementable everywhere under the maintained homophily conditions, with exact relaxed optimality on the stated affine-balanced locus and limiting optimality elsewhere as robustness diverges. The next section turns to parameter regions with strictly positive implementability loss.

4 Positive loss and sharp computation

The preceding section identifies two implementable covariances that can solve the relaxed problem exactly. We now study relaxed points in $\mathcal{E}_m^{\text{blk}}$ lying beyond the covariances attainable by balanced $\{\pm 1\}$ assignment laws. This is the same basic tension that appears in correlation and semidefinite relaxations more broadly: the convex benchmark can contain exposed points beyond the original discrete objects [[Goemans and Williamson, 1995](#), [Deza and Laurent, 1997](#), [Boyd and Vandenberghe, 2004](#)]. In the present two-block design problem, the obstruction is finite and arithmetical. When m is odd, each realized block sum has odd parity, imposing a positive-mass restriction away from the pure within-block spectral direction.

The next result gives a parameter region in which this obstruction is binding. It uses two-block homophily from [assumption 1](#), the low-scale condition from [assumption 3](#), and the odd-block condition from [assumption 4](#). For robustness weights in the stated range, the relevant relaxed covariance is the spread covariance X_{spread} , the vertex with spectral coordinates $(m/(m-1), 0, 0)$. The theorem states that, in a nonempty interval of pseudoinverse weights, this relaxed point is uniquely optimal but not implementable.

Theorem 3. [thm:gap-window] (Positive-Gap Window) *Suppose [assumption 1](#) and [assumption 3](#) hold for the two-block model with block size m and weights a, b . Let*

$$q = 2(m-1), \quad \kappa_{\text{gap}}(m, a, b) = \frac{(2m - a - 3b)(a - b)\sqrt{q}}{a + b},$$

and

$$r_{\text{cut,gap}}(m, a, b, \kappa) = \max \left\{ 0, 2b(a + b) \left(1 - \frac{\kappa}{a - b} \right) \right\}.$$

Define

$$R_x^-(m, a, b, \kappa) = 2b(a + b) \left(1 + \frac{\kappa}{(a - b)\sqrt{q}} \right), \quad R_x^+(m, a, b, \kappa) = (a + b) \left(2m - a - b - \frac{\kappa}{\sqrt{q}} \right),$$

and

$$r_{\text{gap}}^-(m, a, b, \kappa) = \frac{R_x^-(m, a, b, \kappa) + R_x^+(m, a, b, \kappa)}{2}, \quad r_{\text{gap}}^+(m, a, b, \kappa) = R_x^+(m, a, b, \kappa).$$

Then the following two statements hold.

- **(Odd block size.)** If [assumption 4](#) holds, then for every κ satisfying

$$0 \leq \kappa < \kappa_{\text{gap}}(m, a, b),$$

one has

$$r_{\text{cut,gap}}(m, a, b, \kappa) < r_{\text{gap}}^-(m, a, b, \kappa) < r_{\text{gap}}^+(m, a, b, \kappa).$$

Moreover, for every

$$r_{\text{gap}}^-(m, a, b, \kappa) < r < r_{\text{gap}}^+(m, a, b, \kappa),$$

the spread covariance X_{spread} belongs to $\mathcal{E}_m^{\text{blk}}$, and it is the unique strict minimizer of the design objective $F_{r,\kappa}$ from [definition 6](#) over $\mathcal{E}_m^{\text{blk}}$: for every $X \in \mathcal{E}_m^{\text{blk}}$ with $X \neq X_{\text{spread}}$,

$$F_{r,\kappa}(X_{\text{spread}}) < F_{r,\kappa}(X).$$

At the same time,

$$X_{\text{spread}} \notin \mathcal{C}_m^\pm,$$

where $\mathcal{C}_m^\pm$ is the implementable covariance class of [definition 5](#), and the implementability loss of [definition 7](#) satisfies

$$\Delta_m^\pm(r, \kappa) > 0.$$

- **(Even block size.)** If m is even, then every covariance in the block ellipsope is implementable:

$$\mathcal{E}_m^{\text{blk}} \subseteq \mathcal{C}_m^\pm.$$

[theorem 3](#) is the paper’s positive-loss statement. Under its two-block homophily, low-scale, and odd-community hypotheses, the upper bound $\kappa_{\text{gap}}(m, a, b)$ defines a robustness range in which the interval $(r_{\text{gap}}^-, r_{\text{gap}}^+)$ is nonempty and lies strictly beyond the cut-side comparison threshold $r_{\text{cut,gap}}$. Inside that interval, the relaxed optimum is not merely one unattainable minimizer among many; it is the unique strict minimizer over $\mathcal{E}_m^{\text{blk}}$. Since $X_{\text{spread}} \notin \mathcal{C}_m^\pm$, imposing implementability necessarily raises the objective value, giving $\Delta_m^\pm(r, \kappa) > 0$.

The even- m clause marks the opposite parity case. When each block can have zero sign sum, every relaxed covariance in the block ellipsope slice is generated by some block-exchangeable sign law. Thus [theorem 3](#) ties positive loss specifically to the odd-block parity restriction.

The interval result is deliberately local. It identifies one open region where the spread covariance is the relaxed optimum and the loss is strictly positive, but it does not assert a closed-form global boundary for exact attainability outside that region. For computation at arbitrary r and κ , the relevant object is instead the sharp finite active-set formula below.

The formula works in the reduced spectral simplex T_m , obtained from [lemma 2](#): the block symmetry diagonalizes every $X(u, v)$ into three eigendirections, with coordinates $x = 1 - u$ on the within-block contrast space (multiplicity $q = 2(m-1)$), $y = 1 + (m-1)u - mv$ on the between-block contrast direction, and $z_{\text{sp}} = 1 + (m-1)u + mv$ on the all-ones (aggregate) direction; the subscript “sp” marks this last, spread-across-all-units coordinate and distinguishes it from an ambient sign index. In these coordinates $F_{r,\kappa}$ becomes the scalar objective $\phi_{r,\kappa}(x, y, z_{\text{sp}}) = c_x x + c_y y + c_z z_{\text{sp}} + \kappa \sqrt{qx^2 + y^2 + z_{\text{sp}}^2}$, and implementability adds only the parity truncation $d_m \leq y + z_{\text{sp}}$, with $d_m = 0$ for even block sizes and $d_m = 2/m$ for odd block sizes. The theorem’s remaining ingredients are the following finite objects, established in [appendix C](#) ([lemma 7](#) and [lemma 8](#)). For linear coefficients $\alpha^\sharp = (c_x/q, c_y, c_z)$ and norm weights $\beta^\sharp = (1/q, 1, 1)$, the weighted-simplex

objective is $\Phi_{\alpha^\#, \beta^\#, \kappa}(t) = \sum_i \alpha_i^\# t_i + \kappa \sqrt{\sum_i \beta_i^\# t_i^2}$ on the simplex of mass $2m$; an *admissible* active support-multiplier pair (S, λ) is a nonempty $S \subseteq \{x, y, z_{\text{sp}}\}$ and $\lambda \in \mathbb{R}$ with $\sum_{i \in S} (\lambda - \alpha_i^\#)^2 / \beta_i^\# = \kappa^2$, $\alpha_i^\# < \lambda$ on S , and $\lambda \leq \alpha_j^\#$ off S ; the associated *active-set point* $t^{S, \lambda}$ puts mass $2m((\lambda - \alpha_i^\#) / \beta_i^\#) / \sum_{h \in S} (\lambda - \alpha_h^\#) / \beta_h^\#$ on each $i \in S$ and zero off S ; `truncSelector` returns the endpoint or interior optimizer of Φ restricted to the truncation segment $\{y + z_{\text{sp}} = d_m\}$; and $\rho_\star(m, a, b, r, \kappa)$ is the resulting constrained-minus-unconstrained reduced value, i.e. the minimum of $\phi_{r, \kappa}$ over $T_m \cap \{y + z_{\text{sp}} \geq d_m\}$ minus its minimum over T_m . The theorem shows this $\rho_\star$ equals the implementability loss and gives its finite case split.

Theorem 4. [thm:sharp-rho-star] (Sharp Rounding Loss) *Under [assumption 1](#), fix $r, \kappa \in [0, \infty)$. Let*

$$q = 2(m - 1), \quad c_x = q \left((a + b) + \frac{r}{a + b} \right), \quad c_y = 2b + \frac{r}{2b}, \quad c_z = 2m,$$

let

$$T_m = \{(x, y, z_{\text{sp}}) : x, y, z_{\text{sp}} \geq 0, \quad qx + y + z_{\text{sp}} = 2m\},$$

let

$$\phi_{r, \kappa}(x, y, z_{\text{sp}}) = c_x x + c_y y + c_z z_{\text{sp}} + \kappa \sqrt{qx^2 + y^2 + z_{\text{sp}}^2},$$

and let $d_m = 0$ if m is even and $d_m = 2/m$ if m is odd. Then:

- **(Sharp equality and range.)**

$$\rho_\star(m, a, b, r, \kappa) = \Delta_m^\pm(r, \kappa), \quad \rho_\star(m, a, b, r, \kappa) \geq 0, \quad \Delta_m^\pm(r, \kappa) \geq 0,$$

where $\Delta_m^\pm$ is the implementability gap of [definition 7](#).

- **(Zero loss.)**

$$\rho_\star(m, a, b, r, \kappa) = 0$$

if and only if there exists $(x, y, z_{\text{sp}}) \in T_m$ such that

$$\phi_{r, \kappa}(x, y, z_{\text{sp}}) \leq \phi_{r, \kappa}(x', y', z'_{\text{sp}}) \quad \text{for every } (x', y', z'_{\text{sp}}) \in T_m,$$

and

$$d_m \leq y + z_{\text{sp}}.$$

- **(Strict robustness.)** If $\kappa > 0$, then there is a unique relaxed minimizer of $\phi_{r, \kappa}$ over T_m . Moreover,

$$\rho_\star(m, a, b, r, \kappa) = 0$$

if and only if every relaxed minimizer $(x, y, z_{\text{sp}}) \in T_m$ satisfies

$$d_m \leq y + z_{\text{sp}}.$$

- **(Linear case.)** If $\kappa = 0$, then for every $(x, y, z_{\text{sp}}) \in T_m$, the point (x, y, z_{sp}) minimizes $\phi_{r, 0}$ over T_m if and only if

$$x \neq 0 \Rightarrow \frac{c_x}{q} \leq c_y \quad \text{and} \quad \frac{c_x}{q} \leq c_z,$$

$$y \neq 0 \Rightarrow c_y \leq \frac{c_x}{q} \text{ and } c_y \leq c_z,$$

and

$$z_{\text{sp}} \neq 0 \Rightarrow c_z \leq \frac{c_x}{q} \text{ and } c_z \leq c_y.$$

- **(Active-set formula.)** If $\kappa > 0$, then there exist an active support $S \subseteq \{x, y, z_{\text{sp}}\}$ and a multiplier $\lambda \in \mathbb{R}$ such that S and λ are admissible for

$$\alpha^\# = \left(\frac{c_x}{q}, c_y, c_z \right), \quad \beta^\# = \left(\frac{1}{q}, 1, 1 \right), \quad \kappa.$$

Writing $t^{S,\lambda} = (t_x^{S,\lambda}, t_y^{S,\lambda}, t_z^{S,\lambda})$ for the associated active-set point on the simplex of mass $2m$, the reduced point

$$\left(\frac{t_x^{S,\lambda}}{q}, t_y^{S,\lambda}, t_z^{S,\lambda} \right)$$

belongs to T_m , minimizes $\phi_{r,\kappa}$ over T_m , and the relaxed reduced value is

$$2m \lambda.$$

- **(Truncation correction.)** If no relaxed minimizer $(x, y, z_{\text{sp}}) \in T_m$ satisfies $d_m \leq y + z_{\text{sp}}$, then the implementable reduced value equals the weighted-simplex objective with coefficients $\alpha^\#$, weights $\beta^\#$, and robustness weight κ , evaluated at the point on the truncation segment

$$H_{d_m} = \{(2m - d_m, s, d_m - s) : 0 \leq s \leq d_m\}$$

selected by $\text{truncSelector}(2m, d_m, \alpha^\#, \beta^\#, \kappa)$:

$$\text{val}_{\text{impl}}(m, a, b, r, \kappa) = \Phi_{\alpha^\#, \beta^\#, \kappa}(2m - d_m, s_\star, d_m - s_\star),$$

where

$$s_\star = \text{truncSelector}(2m, d_m, \alpha^\#, \beta^\#, \kappa).$$

- **(Even block size.)** If m is even, then

$$\rho_\star(m, a, b, r, \kappa) = 0.$$

[theorem 4](#) turns the implementability question into a finite calculation. The sharp loss $\rho_\star(m, a, b, r, \kappa)$ is exactly the implementability loss $\Delta_m^\pm(r, \kappa)$, so the formula is not an approximation to the rounding penalty. Zero loss occurs precisely when at least one relaxed minimizer satisfies the parity-induced lower bound $d_m \leq y + z_{\text{sp}}$. When $\kappa > 0$, the active-set result yields a unique relaxed minimizer, and the zero-loss condition can be checked at that point alone.

The active-set part supplies the promised finite case split. The coefficients $\alpha^\# = (c_x/q, c_y, c_z)$ and weights $\beta^\# = (1/q, 1, 1)$ convert the reduced objective into a weighted simplex problem over total mass $2m$. For positive robustness, an admissible support S and multiplier λ identify the relaxed minimizer and its value. If that minimizer violates the implementability truncation, the remaining optimization is one-dimensional: the implementable value is obtained on the segment H_{d_m} by the selector appearing in [theorem 4](#). The theorem thereby gives a complete finite procedure for computing $\Delta_m^\pm(r, \kappa)$ for any admissible m, a, b, r, κ .

The two main results in this section therefore play complementary roles. [Theorem 3](#) provides an interpretable open region in which the relaxation has a strictly unattainable optimum for odd block sizes. [Theorem 4](#) gives the exact loss everywhere under [assumption 1](#), including the linear case $\kappa = 0$, the unique-minimizer case $\kappa > 0$, and the even-block case where the loss vanishes. For interference-aware randomized design, the practical implication is a two-step calculation: solve the covariance relaxation, then check its optimizer against the finite sign-assignment restrictions and apply the exact active-set adjustment when needed.

A worked example and parameter map. Take $m = 3$, $a = 2$, $b = 1$, and $\kappa = 0$; these satisfy homophily ($a > b > 0$) and the low-scale regime ($a + 3b = 5 < 2m = 6$). Here $q = 4$, the parity truncation is $d_3 = 2/3$, and the reduced coefficients are $c_x = q((a + b) + r/(a + b)) = 4(3 + r/3)$, $c_y = 2b + r/(2b) = 2 + r/2$, and $c_z = 2m = 6$. The gap-window robustness bound is $\kappa_{\text{gap}} = (2m - a - 3b)(a - b)\sqrt{q}/(a + b) = 2/3$, so $\kappa = 0$ lies in the admissible range. Mapping over $r \geq 0$ then exhibits all three regimes of the theory. For $r < r_{\text{cut}} = \min\{2b(a + b), 2b(2m - 2b)\} = \min\{6, 8\} = 6$, the cut covariance $X_{\text{cut}} = (0, 6, 0)$ is the unique relaxed optimum and is implementable, so $\Delta_3^\pm = 0$ ([theorem 1](#)). For r in the gap window $(r_{\text{gap}}^-, r_{\text{gap}}^+) = (7.5, 9)$, the spread covariance $X_{\text{spread}} = (3/2, 0, 0)$ is the unique relaxed optimum but is *not* implementable ([theorem 3](#)). At $r = 8$, for instance, the unconstrained reduced optimum is at X_{spread} with value $c_x \cdot \frac{3}{2} = 34$, while the implementable optimum lies on the parity slice $y + z_{\text{sp}} = 2/3$ at $(x, y, z_{\text{sp}}) = (4/3, 2/3, 0)$ with value $308/9$; hence

$$\Delta_3^\pm(8, 0) = \frac{308}{9} - 34 = \frac{2}{9} \approx 0.222,$$

a strictly positive but, in this instance, modest price relative to the optimal value 34 (about 0.65%). For even community size the picture collapses: $d_m = 0$, the parity truncation is vacuous, and $\Delta_m^\pm \equiv 0$ throughout. This single example only illustrates the three regimes for one parameter choice; it does not establish that the loss is generally small or generally negligible, and [theorem 4](#) should be used to compute $\Delta_m^\pm$ at any specific (m, a, b, r, κ) of interest. How large the loss can become as a, b, r, κ scale is not settled here and is left open.

[Table 1](#) summarizes the regimes and the result that governs each.

Regime	Relaxed optimum	Implementable?	Loss $\Delta_m^\pm$
Cut ($r < r_{\text{cut}}$)	$X_{\text{cut}} = (0, 2m, 0)$	yes (via P_{cut})	0 (Thm theorem 1)
Robust locus ($a + 3b = 2m$, $r = 2b(a + b)$)	$I_n = (1, 1, 1)$	yes (via P_{iid})	0 (Thm theorem 2)
Odd-parity window (m odd, low-scale)	$X_{\text{spread}} = (\frac{m}{m-1}, 0, 0)$	no	> 0 (Thm theorem 3)
Even m	—	every $X \in \mathcal{E}_m^{\text{blk}}$	0 (Thm theorem 4)
General (m, a, b, r, κ)	active-set point	iff parity slice met	$\rho_\star$ (Thm theorem 4)

Table 1: Regimes of the two-block implementability problem. Coordinates are the reduced spectral coordinates (x, y, z_{sp}) ; “low-scale” abbreviates $a + 3b < 2m$ with the robustness range of [theorem 3](#).

5 Discussion and extensions

Relation to covariance-based network design. The design criterion and ellipsope relaxation build on covariance-based experimental design for network interference, particularly the

semidefinite formulation of [Thiyageswaran et al. \[2026\]](#), and on the broader ellipsope-versus-cut-polytope literature [[Goemans and Williamson, 1995](#), [Deza and Laurent, 1997](#), [Boyd and Vandenberghe, 2004](#)]. This paper contributes an exact implementability theory for the homophilous two-block family: a joint symmetry reduction of relaxed covariances and implementable laws, an odd-parity obstruction producing positive loss, and a finite active-set correction computing $\Delta_m^\pm$ exactly.

Viewed abstractly, $\mathcal{C}_m^\pm$ is a moment set for block-exchangeable Rademacher vectors and a two-coordinate projection of the correlation polytope. The paper pins down the exact boundary of this low-dimensional exchangeable slice through the odd-block parity truncation $y + z_{\text{sp}} \geq 2/m$, then converts that boundary into the exact rounding loss. Future work can connect this family-specific characterization more fully to general correlation-polytope and exchangeable-moment results.

The results above give a finite-sample account of when a covariance relaxation can be read literally as a randomized design. In the two-block homophily model, the relaxed block ellipsope $\mathcal{E}_m^{\text{blk}}$ serves both as a computational device and as a benchmark for exact implementability by a sign-symmetric, block-exchangeable $\{\pm 1\}$ assignment law. [Theorem 1](#) gives a region in which the relaxed optimum is generated by an assignment law, whereas [theorem 3](#) gives a region with a uniquely optimal relaxed covariance and strictly positive implementability loss.

For experimental design under interference, the covariance matrix is a property of the assignment mechanism, and implementation requires a covariance induced by a randomization law. The implementability loss $\Delta_m^\pm(r, \kappa)$ therefore has a direct design interpretation: it is the exact objective-value price of returning from the relaxed covariance problem to feasible randomized assignment. [Theorem 4](#) computes that price through a finite active-set calculation.

The iid result points in the opposite direction. Independent fair assignment is always available as a randomized design and has a familiar role in design-based inference. [Theorem 2](#) identifies exact finite-robustness optimality on the affine-balanced locus $a + 3b = 2m$ and $r = 2b(a + b)$, while relaxed minimizers elsewhere converge to I_n as κ grows. Thus robustness rationalizes iid assignment either as an exact solution on the locus or as a limiting target elsewhere.

The positive-loss result is arithmetic. For odd m , each realized block sum has absolute value at least one. Under the hypotheses and robustness range stated in [theorem 3](#), this parity restriction binds and places the uniquely optimal spread covariance outside $\mathcal{C}_m^\pm$. For even m , every relaxed block covariance is implementable in the maintained symmetric model. The theorem therefore isolates odd-block parity as the source of positive loss.

5.1 Limitations and future work

The two-block structure is deliberately stylized, but it captures a common empirical feature of networked populations: ties and exposures are often stronger within social groups than between them. Homophily and community structure are central themes in social network analysis [[McPherson et al., 2001](#), [Newman, 2002](#)], and stochastic block models provide a canonical probabilistic language for such clustered networks [[Holland et al., 1983](#), [Abbe, 2018](#)]. Field and policy experiments with interference often face analogous clustering through villages, schools, households, online neighborhoods, or other exposure groups [[Banerjee et al., 2013](#), [Baird et al., 2018](#), [Ugander and Yin, 2020](#), [Li et al., 2021](#)]. The present model abstracts from those empirical details in order to isolate an exact design question: after symmetry reduction, which relaxed covariance matrices are generated by actual sign assignments?

Several extensions follow naturally, but they require additional arguments beyond the state-

ments in this paper. The present results do not cover unequal block sizes, where both the spectral reduction and the block-sum lattice would require new arguments. The present results also do not cover more than two communities; the three-coordinate reduction (x, y, z_{sp}) and the implementability characterization would require modification. In those settings, one should not expect the parity correction d_m or the active-set formula in [theorem 4](#) to carry over without modification.

A second extension concerns additional restrictions on the randomization scheme. The class $\mathcal{P}_m^{\text{sym}}$ imposes sign symmetry and block exchangeability, but it does not require a fixed treated count in every realization, nor does it impose covariate balance, rerandomization acceptance rules, or cluster-level assignment constraints. Such restrictions are common in empirical practice and can be important for finite-sample precision and credibility [[Sävje, 2021](#), [Zigler and Papadogeorgou, 2018](#)]. They would shrink the implementable covariance class and could introduce losses even in cases where $\mathcal{E}_m^{\text{blk}} \subseteq \mathcal{C}_m^{\pm}$ under the present assumptions.

A third issue is that the graph itself is treated as fixed. In applications, measured networks may be incomplete, exposure weights may be estimated, and the relevant interference neighborhood may be uncertain. The objective $F_{r,\kappa}$ can be interpreted conditionally on the chosen graph weights, but the results here do not address robustness to network measurement error or to alternative exposure mappings. Those questions are important for empirical deployment because different plausible graphs can imply different design priorities, especially when homophily is strong.

Finally, the analysis is finite-sample throughout. The exact formulas are indexed by m, a, b, r, κ , and the odd-block obstruction is stated at the finite block size used by the design. Although the parity truncation becomes numerically smaller as m grows, the consequences for objective values depend on the full set of coefficients in [theorem 4](#). Any asymptotic simplification should therefore be derived from the finite active-set expression rather than assumed from the geometry alone.

Appendices

A Reduced geometry and proof ingredients

This appendix makes visible the auxiliary geometry behind the two main optimization statements: the reduced coordinates, vertex certificates, parity restriction, and finite active-set calculation used in [theorem 3](#) and [theorem 4](#). The reduction is a low-dimensional convex-analytic one: the block symmetry turns the covariance problem into a weighted simplex problem, in the same general spirit as ellipotope and semidefinite relaxations for discrete optimization [[Goemans and Williamson, 1995](#), [Deza and Laurent, 1997](#), [Boyd and Vandenberghe, 2004](#)].

The first ingredient identifies the spectral coordinates in which the block ellipotope becomes a simplex and the objective becomes a scalar function.

Lemma 2. [[lem:block-spectral-coordinates](#)] (Block Spectral Coordinates) *For any real numbers r, κ, u, v , assume:*

- (*Homophily.*) [assumption 1](#) holds for the block size m and weights a, b .
- (*Spectral coordinates.*) Set

$$q = 2(m - 1), \quad x = 1 - u, \quad y = 1 + (m - 1)u - mv, \quad z_{\text{sp}} = 1 + (m - 1)u + mv.$$

- *(Reduced coefficients.)* Set

$$c_x = q \left((a+b) + \frac{r}{a+b} \right), \quad c_y = 2b + \frac{r}{2b}, \quad c_z = 2m.$$

Then the block covariance matrix $X(u, v)$ belongs to the block elliptope $\mathcal{E}_m^{\text{blk}}$ if and only if

$$x \geq 0, \quad y \geq 0, \quad z_{\text{sp}} \geq 0, \quad qx + y + z_{\text{sp}} = 2m.$$

Moreover, for the design objective of [definition 6](#),

$$F_{r,\kappa}(X(u, v)) = c_x x + c_y y + c_z z_{\text{sp}} + \kappa \sqrt{qx^2 + y^2 + z_{\text{sp}}^2}.$$

Finally,

$$X_{\text{cut}} = X(1, -1) \quad \text{and} \quad (1 - 1, 1 + (m-1) \cdot 1 - m(-1), 1 + (m-1) \cdot 1 + m(-1)) = (0, 2m, 0),$$

while

$$I_n = X(0, 0) \quad \text{and} \quad (1 - 0, 1 + (m-1) \cdot 0 - m \cdot 0, 1 + (m-1) \cdot 0 + m \cdot 0) = (1, 1, 1).$$

Proof of [lemma 2](#). [assumption 1](#) gives $m \geq 2$, so the block-size normalizations used below are nonzero. For $X(u, v)$, the block-elliptope inequalities are precisely the three nonnegativity conditions

$$1 - u \geq 0, \quad 1 + (m-1)u - mv \geq 0, \quad 1 + (m-1)u + mv \geq 0,$$

and the coordinates satisfy the trace identity

$$q(1 - u) + (1 + (m-1)u - mv) + (1 + (m-1)u + mv) = 2m.$$

Thus block-elliptope membership is equivalent to $(x, y, z_{\text{sp}}) \in T_m$.

For the objective formula, compute the four terms on $X(u, v)$. Pair counting gives

$$\text{tr}(J_n X(u, v)) = 2m z_{\text{sp}}, \quad \|X(u, v)\|_{S_2} = \sqrt{qx^2 + y^2 + z_{\text{sp}}^2}.$$

The projections onto the all-ones and block-sign directions give traces z_{sp} and y , respectively, and the diagonal trace is $2m$. From the two-block Laplacian and its spectral-action pseudoinverse, this yields

$$\text{tr}(L_m X(u, v)) = q(a+b)x + 2by, \quad \text{tr}(L_m^\dagger X(u, v)) = \frac{qx}{a+b} + \frac{y}{2b}.$$

Substituting these four identities into [definition 6](#) gives

$$F_{r,\kappa}(X(u, v)) = c_x x + c_y y + c_z z_{\text{sp}} + \kappa \sqrt{qx^2 + y^2 + z_{\text{sp}}^2}.$$

Finally, the cut covariance is the block matrix with within-block entries 1 and across-block entries -1 , hence $X_{\text{cut}} = X(1, -1)$, and direct substitution gives $(0, 2m, 0)$. Likewise the identity matrix is $X(0, 0)$, and direct substitution gives $(1, 1, 1)$. $\square$

[lemma 2](#) is the coordinate change used throughout the main text. It replaces the two covariance parameters u, v by nonnegative spectral coordinates satisfying one affine constraint. In these coordinates, the cut covariance is the vertex $(0, 2m, 0)$, the iid covariance is the center-like point $(1, 1, 1)$, and the objective separates into a linear part plus a weighted Euclidean norm.

The next two statements are vertex certificates. They give simple coefficient inequalities under which either the cut vertex or the spread vertex is the unique minimizer of the reduced problem.

Lemma 3. [lem:cut-vertex-certificate] (Cut Vertex Certificate) *Let m be fixed and write $q = 2(m - 1)$. Suppose that*

- *(Positive spectral multiplicity.)* $q > 0$.
- *(Nonnegative robustness.)* $\kappa \geq 0$.
- *(Cut-side coefficient separation.)* The coefficients satisfy

$$\frac{c_x}{q} > c_y + \kappa \quad \text{and} \quad c_z > c_y + \kappa.$$

Then $(0, 2m, 0) \in T_m$, where

$$T_m = \{(x, y, z_{\text{sp}}) : x \geq 0, y \geq 0, z_{\text{sp}} \geq 0, qx + y + z_{\text{sp}} = 2m\}.$$

Moreover, for every $(x, y, z_{\text{sp}}) \in T_m$ with $(x, y, z_{\text{sp}}) \neq (0, 2m, 0)$,

$$c_x \cdot 0 + c_y(2m) + c_z \cdot 0 + \kappa \sqrt{q \cdot 0^2 + (2m)^2 + 0^2} < c_x x + c_y y + c_z z_{\text{sp}} + \kappa \sqrt{qx^2 + y^2 + z_{\text{sp}}^2}.$$

Thus $(0, 2m, 0)$ is the unique minimizer on T_m of the reduced objective.

Proof of lemma 3. Write $q = 2(m - 1)$ and $M = 2m$. The point $(0, M, 0)$ has nonnegative coordinates and satisfies $q \cdot 0 + M + 0 = M$, so it belongs to T_m .

Now take $(x, y, z_{\text{sp}}) \in T_m$, different from $(0, M, 0)$. Since $q > 0$, $x, z_{\text{sp}} \geq 0$, and $qx + y + z_{\text{sp}} = M$, we have

$$y \leq \sqrt{qx^2 + y^2 + z_{\text{sp}}^2},$$

and hence $\kappa y \leq \kappa \sqrt{qx^2 + y^2 + z_{\text{sp}}^2}$. The coefficient assumptions imply

$$c_x - qc_y - q\kappa > 0, \quad c_z - c_y - \kappa > 0.$$

If both $x = 0$ and $z_{\text{sp}} = 0$, the simplex equation would force $y = M$, contradicting the assumption that the point is not $(0, M, 0)$. Therefore at least one of x, z_{sp} is positive, and so

$$0 < (c_x - qc_y - q\kappa)x + (c_z - c_y - \kappa)z_{\text{sp}}.$$

Using $M = qx + y + z_{\text{sp}}$, this is exactly

$$c_y M + \kappa M < c_x x + c_y y + c_z z_{\text{sp}} + \kappa y.$$

Combining this strict inequality with the norm lower bound above gives the stated strict comparison of reduced objectives. Hence no point of T_m other than $(0, 2m, 0)$ can attain the same value, so the cut vertex is the unique minimizer. $\square$

[lemma 3](#) is the reduced certificate underlying the cut exactness result. The coefficient separation says that moving mass away from the y -coordinate toward either the within-block spectral coordinate or the aggregate coordinate is too costly, even after accounting for the robustness term.

Lemma 4. `[lem:spread-vertex-certificate]` (Spread Vertex Certificate) *Let $m \in \mathbb{N}$, let $c_x, c_y, c_z, \kappa \in \mathbb{R}$, and set $q = 2(m - 1)$. Suppose that*

- *(Positive multiplicity.)* $q > 0$.
- *(Nonnegative robustness.)* $\kappa \geq 0$.
- *(The y -coefficient is large.)* $c_y > c_x/q + \kappa/\sqrt{q}$.
- *(The z_{sp} -coefficient is large.)* $c_z > c_x/q + \kappa/\sqrt{q}$.

Define

$$T_m = \{(x, y, z_{\text{sp}}) : x \geq 0, y \geq 0, z_{\text{sp}} \geq 0, qx + y + z_{\text{sp}} = 2m\}$$

and

$$\phi(x, y, z_{\text{sp}}) = c_x x + c_y y + c_z z_{\text{sp}} + \kappa \sqrt{qx^2 + y^2 + z_{\text{sp}}^2}.$$

Then $(2m/q, 0, 0) \in T_m$, and for every $(x, y, z_{\text{sp}}) \in T_m$ with $(x, y, z_{\text{sp}}) \neq (2m/q, 0, 0)$,

$$\phi(2m/q, 0, 0) < \phi(x, y, z_{\text{sp}}).$$

Proof of lemma 4. Put $q = 2(m - 1)$, $M = 2m$, and $s = \sqrt{q}$. Since $q > 0$, $s > 0$. The point $(M/q, 0, 0)$ has nonnegative coordinates and satisfies $q(M/q) + 0 + 0 = M$, so it lies in T_m .

Let $(x, y, z_{\text{sp}}) \in T_m$ be different from $(M/q, 0, 0)$. From $s^2 = q$ we have

$$(sx)^2 \leq qx^2 + y^2 + z_{\text{sp}}^2,$$

hence

$$\kappa sx \leq \kappa \sqrt{qx^2 + y^2 + z_{\text{sp}}^2}.$$

Also

$$\sqrt{q(M/q)^2} = s(M/q).$$

The assumptions on c_y and c_z , after multiplication by $s > 0$, give

$$(c_y - c_x/q)s - \kappa > 0, \quad (c_z - c_x/q)s - \kappa > 0.$$

If $y = z_{\text{sp}} = 0$, the simplex equation forces $x = M/q$, contrary to the assumed inequality of points; hence y or z_{sp} is positive. Therefore

$$0 < ((c_y - c_x/q)s - \kappa)y + ((c_z - c_x/q)s - \kappa)z_{\text{sp}}.$$

Using $M = qx + y + z_{\text{sp}}$, the algebraic identity

$$\begin{aligned} & (c_x x + c_y y + c_z z_{\text{sp}} + \kappa sx - c_x(M/q) - \kappa s(M/q))s \\ &= ((c_y - c_x/q)s - \kappa)y + ((c_z - c_x/q)s - \kappa)z_{\text{sp}} \end{aligned}$$

then implies

$$c_x(M/q) + \kappa s(M/q) < c_x x + c_y y + c_z z_{\text{sp}} + \kappa s x.$$

Together with the norm lower bound, this is exactly

$$\phi(M/q, 0, 0) < \phi(x, y, z_{\text{sp}}).$$

□

[lemma 4](#) plays the corresponding role for the spread vertex. In the notation of [theorem 3](#), this vertex is the reduced representation of X_{spread} . The coefficient inequalities make the x -direction uniquely cheapest, so the relaxed optimizer concentrates all simplex mass there.

The iid result uses a different geometric fact: the point $(1, 1, 1)$ is the unique minimizer of the Frobenius part on the simplex, but it minimizes the full objective only when the linear coefficients are balanced across the three spectral directions.

Lemma 5. [`lem:frobenius-center-certificate`] (Frobenius Center Certificate) *Let $m \in \mathbb{N}$, set $q = 2(m - 1)$, and suppose $q > 0$. For any real coefficients c_x, c_y, c_z and any real κ , define*

$$T_m = \{(x, y, z_{\text{sp}}) : x, y, z_{\text{sp}} \geq 0, qx + y + z_{\text{sp}} = 2m\}$$

and

$$\phi_{r,\kappa}(x, y, z_{\text{sp}}) = c_x x + c_y y + c_z z_{\text{sp}} + \kappa \sqrt{qx^2 + y^2 + z_{\text{sp}}^2}.$$

Then:

- **(Frobenius center.)** *The point $(1, 1, 1)$ belongs to T_m , and for every $(x, y, z_{\text{sp}}) \in T_m$ with $(x, y, z_{\text{sp}}) \neq (1, 1, 1)$,*

$$\sqrt{q \cdot 1^2 + 1^2 + 1^2} < \sqrt{qx^2 + y^2 + z_{\text{sp}}^2}.$$

- **(Certificate necessity.)** *If $(1, 1, 1)$ minimizes $\phi_{r,\kappa}$ over T_m , that is, if*

$$\phi_{r,\kappa}(1, 1, 1) \leq \phi_{r,\kappa}(x, y, z_{\text{sp}}) \quad \text{for all } (x, y, z_{\text{sp}}) \in T_m,$$

then

$$\frac{c_x}{q} = c_y \quad \text{and} \quad c_y = c_z.$$

- **(Strict optimality.)** *If $\kappa > 0$, $\frac{c_x}{q} = c_y$, and $c_y = c_z$, then for every $(x, y, z_{\text{sp}}) \in T_m$ with $(x, y, z_{\text{sp}}) \neq (1, 1, 1)$,*

$$\phi_{r,\kappa}(1, 1, 1) < \phi_{r,\kappa}(x, y, z_{\text{sp}}).$$

Proof of lemma 5. Let $q = 2(m - 1)$. Since $q + 2 = 2m$, the point $(1, 1, 1)$ has nonnegative coordinates and satisfies $q \cdot 1 + 1 + 1 = 2m$, hence belongs to T_m . For any $(x, y, z_{\text{sp}}) \in T_m$, the simplex equation gives

$$qx^2 + y^2 + z_{\text{sp}}^2 = (q + 2) + q(x - 1)^2 + (y - 1)^2 + (z_{\text{sp}} - 1)^2.$$

If the point is not $(1, 1, 1)$, the last three-square sum is strictly positive, since $q > 0$. Taking square roots gives the stated strict Frobenius-center inequality.

Assume next that $(1, 1, 1)$ minimizes $\phi_{r,\kappa}$ on T_m . Along the feasible tangent curve

$$t \mapsto (1 + t, 1 - qt, 1)$$

for all sufficiently small t , the point $t = 0$ is a local minimum. The Frobenius term has derivative 0 at $t = 0$, so the derivative of the full objective there is $c_x - qc_y$; local minimality forces $c_x - qc_y = 0$, equivalently $c_x/q = c_y$. Similarly, along

$$t \mapsto (1, 1 + t, 1 - t),$$

local minimality at $t = 0$ and the vanishing first derivative of the Frobenius term give derivative $c_y - c_z$, hence $c_y = c_z$.

Finally suppose $\kappa > 0$, $c_x/q = c_y$, and $c_y = c_z$. Then $c_x = qc_y$, and for every point of T_m ,

$$c_x \cdot 1 + c_y \cdot 1 + c_z \cdot 1 = c_x x + c_y y + c_z z_{\text{sp}},$$

because both sides equal $c_y(qx + y + z_{\text{sp}}) = c_y(q + 2)$. Thus the linear part of the objective is constant on T_m . For every point other than $(1, 1, 1)$, the first part gives a strictly larger Frobenius norm, and multiplication by $\kappa > 0$ preserves strictness. Therefore $(1, 1, 1)$ is a strict minimizer. $\square$

[lemma 5](#) explains why the iid covariance has a knife-edge exactness condition in [theorem 2](#). The norm term alone favors the point $(1, 1, 1)$, but the linear coefficients must be equal after accounting for the q -multiplicity of the x -coordinate. When that equality fails, the linear part tilts the objective away from the iid point at every finite positive robustness weight.

The next statement is the implementability restriction in reduced form. It is where the parity of the block size enters the geometry.

Lemma 6. [[lem:pm-reduced-slice-characterization](#)] (Parity-Truncated Reduced Slice)
Let $m \geq 2$ be an integer and let $u, v \in \mathbb{R}$. Write

$$x = 1 - u, \quad y = 1 + (m - 1)u - mv, \quad z_{\text{sp}} = 1 + (m - 1)u + mv,$$

and let $d_m = 0$ when m is even and $d_m = 2/m$ when m is odd. Then the block covariance matrix $X(u, v)$ belongs to the implementable covariance class $\mathcal{C}_m^\pm$ of [definition 5](#) if and only if

$$x \geq 0, \quad y \geq 0, \quad z_{\text{sp}} \geq 0, \quad 2(m - 1)x + y + z_{\text{sp}} = 2m,$$

and

$$y + z_{\text{sp}} \geq d_m.$$

Proof of lemma 6. We prove the two implications separately. Suppose first that $X(u, v) \in \mathcal{C}_m^\pm$. By [definition 5](#), there is a block-exchangeable sign-symmetric design whose second moment is $X(u, v)$. Reading off the entries of this second moment and applying the reduced-slice necessity calculation gives

$$x \geq 0, \quad y \geq 0, \quad z_{\text{sp}} \geq 0, \quad 2(m - 1)x + y + z_{\text{sp}} = 2m,$$

and also the parity lower bound $d_m \leq y + z_{\text{sp}}$. In that calculation, the nonnegativity of y and z_{sp} comes from the second moments of the block-sum combinations $S_A - S_B$ and $S_A + S_B$, while for odd m the parity of the block sums gives $S_A^2, S_B^2 \geq 1$ and hence $y + z_{\text{sp}} \geq 2/m$; for even m the

threshold is 0. Conversely, assume the displayed reduced-triangle conditions and $d_m \leq y + z_{\text{sp}}$. The sufficiency construction realizes the point by a block-exchangeable sign-symmetric design: for even m it is the barycentric mixture of the cut, all-equal, and spread vertex designs; for odd m the same reduced-triangle point is realized inside the parity-truncated quadrilateral cut out by $y + z_{\text{sp}} \geq 2/m$. Thus $X(u, v)$ is the covariance of a design in $\mathcal{P}_m^{\text{sym}}$, so $X(u, v) \in \mathcal{C}_m^\pm$. Combining the two directions proves the characterization. $\square$

[lemma 6](#) gives implementability a transparent low-dimensional form. Relative to the relaxed simplex, the implementable slice adds the lower bound $y + z_{\text{sp}} \geq d_m$. For even m , $d_m = 0$, so this lower bound is already implied by nonnegativity. For odd m , the positive lower bound records the positive squared magnitude of the realized block sums.

It remains to solve the reduced optimization problem. The next lemma gives the unconstrained weighted-simplex solution as a finite active-set calculation.

Lemma 7. [lem:weighted-simplex-active-set] (Weighted Simplex Active Set) *Let $m \geq 2$, let $M > 0$ satisfy $M = 2m$, and let $\mathcal{I} = \{x, y, z_{\text{sp}}\}$. Let $\alpha = (\alpha_i)_{i \in \mathcal{I}}$, let $\kappa \geq 0$, and let $\beta = (\beta_i)_{i \in \mathcal{I}}$ have strictly positive entries with*

$$\beta_x = \frac{1}{2(m-1)}, \quad \beta_y = \beta_{z_{\text{sp}}} = 1.$$

For $t \in \mathbb{R}^{\mathcal{I}}$, write

$$\Delta_M = \left\{ t : t_i \geq 0 \text{ for all } i \in \mathcal{I}, \sum_{i \in \mathcal{I}} t_i = M \right\},$$

and

$$\Phi(t) = \sum_{i \in \mathcal{I}} \alpha_i t_i + \kappa \sqrt{\sum_{i \in \mathcal{I}} \beta_i t_i^2}.$$

Then the following hold.

- **(Positive robustness.)** *If $\kappa > 0$, there is a unique pair (S, λ) , where $S \subseteq \mathcal{I}$ is nonempty and $\lambda \in \mathbb{R}$, such that*

$$\sum_{i \in S} \frac{(\lambda - \alpha_i)^2}{\beta_i} = \kappa^2, \quad \alpha_i < \lambda \text{ for every } i \in S, \quad \lambda \leq \alpha_j \text{ for every } j \notin S.$$

For this unique pair, define

$$t_i^* = \begin{cases} M \frac{(\lambda - \alpha_i)/\beta_i}{\sum_{h \in S} (\lambda - \alpha_h)/\beta_h}, & i \in S, \\ 0, & i \notin S. \end{cases}$$

Then $t^* \in \Delta_M$, and t^* is the unique minimizer of Φ over Δ_M in the strict sense that, for every $s \in \Delta_M$ with $s \neq t^*$,

$$\Phi(t^*) < \Phi(s).$$

Its value is

$$\Phi(t^*) = M\lambda.$$

- **(Zero robustness.)** If $\kappa = 0$, then for every $t \in \mathbb{R}^{\mathcal{I}}$,

$$\left(t \in \Delta_M \text{ and } \Phi(t) \leq \Phi(s) \text{ for every } s \in \Delta_M \right) \iff t \in \Delta_M \text{ and whenever } t_i \neq 0, \alpha_i \leq \alpha_j \text{ for every } j \in \mathcal{I}$$

Equivalently, the minimizer set is exactly the exposed α -minimizing face of Δ_M .

Proof of lemma 7. First assume $\kappa > 0$. The admissible active-set data exist by applying the intermediate value argument to the increasing threshold function

$$\lambda \mapsto \sum_{i \in \mathcal{I}} \frac{\max\{\lambda - \alpha_i, 0\}^2}{\beta_i};$$

one obtains a nonempty support $S = \{i : \alpha_i < \lambda\}$ and a multiplier λ satisfying exactly the displayed admissibility equations and inequalities. For this pair, the denominator $\sum_{h \in S} (\lambda - \alpha_h)/\beta_h$ is positive, so the displayed formula defines a nonnegative vector whose coordinates sum to M ; hence $t^* \in \Delta_M$. The KKT certificate for this point gives strict optimality on the simplex. Namely, on S the stationarity relation identifies $\lambda - \alpha_i$ with the weighted-norm gradient, while outside S the inequalities $\lambda \leq \alpha_j$ and $s_j \geq 0$ give the complementary nonnegative remainder. Strict Cauchy–Schwarz for any $s \neq t^*$ then yields $\Phi(t^*) < \Phi(s)$. The same stationarity identity, multiplied by t_i^* and summed over i , gives

$$\Phi(t^*) = \lambda \sum_i t_i^* = M\lambda.$$

It remains in the positive- κ case to see uniqueness of (S, λ) . If another admissible pair (S', λ') had the stated properties, its active-set point would also be a strict minimizer, so the two points must coincide; otherwise each would have strictly smaller value than the other. The support is recovered from the point by the equivalence $i \in S$ if and only if $t_i^* > 0$, so $S = S'$. Finally the value identity gives $M\lambda = M\lambda'$, and $M > 0$ implies $\lambda = \lambda'$.

Now assume $\kappa = 0$. Then $\Phi(t) = \sum_i \alpha_i t_i$ is linear. Choose a coordinate k with minimal α_k . If t is a minimizer, comparison with the vertex Me_k gives

$$\sum_i (\alpha_i - \alpha_k) t_i \leq 0,$$

while each summand is nonnegative; hence every nonzero coordinate of t lies among the α -minimizing coordinates. Conversely, any simplex point supported only on those minimizing coordinates has value $M\alpha_k$, and every simplex point has value at least $M\alpha_k$. This is exactly the exposed α -minimizing face. $\square$

lemma 7 is the finite case split behind the active-set formula in theorem 4. For $\kappa > 0$, the support and multiplier determine both the optimizer and the value. For $\kappa = 0$, the norm term drops out and the minimizers form the exposed face associated with the smallest linear coefficient.

The implementable problem may add the parity lower bound. The following truncation statement records how the solution changes when the unconstrained active-set point lies outside that truncated simplex.

Lemma 8. [lem:weighted-simplex-truncation] (Weighted Simplex Truncation) *Let $m \geq 2$, let $M, d \in \mathbb{R}$, and let $\alpha = (\alpha_x, \alpha_y, \alpha_z)$, $\beta = (\beta_x, \beta_y, \beta_z)$, and $\kappa \in \mathbb{R}$. Suppose:*

- **(Mass and truncation.)** $M > 0$, $M = 2m$, and $0 \leq d \leq M$.
- **(Weights.)** Each β_i is strictly positive and

$$\beta = \left(\frac{1}{2(m-1)}, 1, 1 \right),$$

so in particular $\beta_y = \beta_z = 1$.

- **(Robustness.)** $\kappa \geq 0$.

Write

$$K_d = \{t \in \Delta_M : t_y + t_z \geq d\}, \quad \Phi(t) = \sum_i \alpha_i t_i + \kappa \sqrt{\sum_i \beta_i t_i^2}.$$

For $\kappa > 0$, for every nonempty active support $S \subseteq \{x, y, z\}$ and multiplier λ satisfying

$$\sum_{i \in S} \frac{(\lambda - \alpha_i)^2}{\beta_i} = \kappa^2, \quad \alpha_i < \lambda \text{ for } i \in S, \quad \lambda \leq \alpha_j \text{ for } j \notin S,$$

let t^{rel} be the active-set point

$$t_i^{\text{rel}} = \begin{cases} \frac{M((\lambda - \alpha_i)/\beta_i)}{\sum_{h \in S} (\lambda - \alpha_h)/\beta_h}, & i \in S, \\ 0, & i \notin S. \end{cases}$$

If $t^{\text{rel}} \in K_d$, then

$$\Phi(t^{\text{rel}}) \leq \Phi(s) \quad \text{for every } s \in K_d.$$

If $t_y^{\text{rel}} + t_z^{\text{rel}} < d$, then

$$t^* = (M - d, s^*, d - s^*) \in K_d$$

and

$$\Phi(t^*) \leq \Phi(s) \quad \text{for every } s \in K_d,$$

where $s^* = \text{truncSelector}(M, d, \alpha, \beta, \kappa)$, namely with

$$\delta = \alpha_y - \alpha_z, \quad A = \beta_x(M - d)^2,$$

$$s^* = \begin{cases} 0, & \delta \geq \frac{\kappa d}{\sqrt{A + d^2}}, \\ d, & \delta \leq -\frac{\kappa d}{\sqrt{A + d^2}}, \\ \frac{d - \delta \sqrt{(A + d^2/2)/(\kappa^2 - \delta^2/2)}}{2}, & \text{otherwise.} \end{cases}$$

For $\kappa = 0$, let t^{rel} be a point of the exposed α -minimizing face

$$\{t \in \Delta_M : t_i \neq 0 \Rightarrow \alpha_i \leq \alpha_j \text{ for every } j\}.$$

Assume the tie convention that, if this exposed face contains at least one point of K_d , then the selected point t^{rel} lies in K_d . If $t^{\text{rel}} \in K_d$, then

$$\Phi(t^{\text{rel}}) \leq \Phi(s) \quad \text{for every } s \in K_d.$$

If $t_y^{\text{rel}} + t_z^{\text{rel}} < d$, then

$$t^* = (M - d, s^*, d - s^*) \in K_d$$

and

$$\Phi(t^*) \leq \Phi(s) \quad \text{for every } s \in K_d,$$

where $s^* = \text{truncSelector}(M, d, \alpha, \beta, 0)$, equivalently, in the ordered selector convention, $s^* = 0$ if $\alpha_y - \alpha_z \geq 0$ and $s^* = d$ if $\alpha_y - \alpha_z < 0$.

Proof of lemma 8. We first record the truncation dichotomy used in both robustness regimes. Let $t^{\text{rel}} \in \Delta_M$ minimize Φ over the full simplex. If $t^{\text{rel}} \in K_d$, then every $s \in K_d$ is also in Δ_M , so the full-simplex minimality immediately gives $\Phi(t^{\text{rel}}) \leq \Phi(s)$. If instead $t_y^{\text{rel}} + t_z^{\text{rel}} < d$, then $d > 0$. The selector $s^* = \text{truncSelector}(M, d, \alpha, \beta, \kappa)$ lies in $[0, d]$, so

$$(M - d, s^*, d - s^*) \in K_d.$$

For any $s \in K_d$, convexity of the objective along the segment from s to t^{rel} shows that s is dominated by some point on the boundary segment

$$H_d = \{(M - d, \sigma, d - \sigma) : 0 \leq \sigma \leq d\}.$$

On this segment the objective reduces to the one-dimensional convex function

$$\alpha_x(M - d) + \alpha_y\sigma + \alpha_z(d - \sigma) + \kappa\sqrt{\beta_x(M - d)^2 + \sigma^2 + (d - \sigma)^2},$$

and the displayed selector is chosen exactly so that its tangent inequality holds on $[0, d]$. Therefore the selected boundary point minimizes the objective over H_d , and hence over all of K_d .

When $\kappa > 0$, the assumed admissibility of (S, λ) gives that the displayed active-set point t^{rel} lies in Δ_M and minimizes Φ over the full simplex. Applying the dichotomy just proved gives the two asserted alternatives: if $t^{\text{rel}} \in K_d$, it is already optimal on K_d ; if $t_y^{\text{rel}} + t_z^{\text{rel}} < d$, the selected point $(M - d, s^*, d - s^*)$ is feasible and optimal on K_d .

When $\kappa = 0$, a point t^{rel} of the exposed α -minimizing face minimizes the linear objective over Δ_M . The same dichotomy applies verbatim. Thus, if the selected relaxed point lies in K_d , it is optimal over K_d ; if its $y + z$ mass is below d , the constrained optimum lies on H_d at

$$s^* = \text{truncSelector}(M, d, \alpha, \beta, 0).$$

For $\kappa = 0$, the selector guards reduce to the ordered endpoint rule: $s^* = 0$ when $\alpha_y - \alpha_z \geq 0$, and $s^* = d$ when $\alpha_y - \alpha_z < 0$. $\square$

lemma 8 reduces the constrained implementable calculation to a one-dimensional boundary segment whenever the relaxed optimizer violates the lower bound. The selector chooses either an endpoint of that segment or the unique interior point identified by the displayed expression. This is the source of the truncation correction in theorem 4.

The final ingredient ties the reduced optimization problems back to the implementability loss defined in the main text.

Lemma 9. [lem:rounding-gap-reduction] (Reduced Gap Identity) *For any real r and any $\kappa \geq 0$, under assumption 1, the implementability loss of definition 7 satisfies*

$$\Delta_m^\pm(r, \kappa) = \inf_{\substack{(x, y, z_{\text{sp}}) \in T_m \\ y + z_{\text{sp}} \geq d_m}} \phi_{r, \kappa}(x, y, z_{\text{sp}}) - \inf_{(x, y, z_{\text{sp}}) \in T_m} \phi_{r, \kappa}(x, y, z_{\text{sp}}).$$

Proof of lemma 9. Set $q = 2(m - 1)$. Under assumption 1 we have $m \geq 2$. The change of variables is invertible on the reduced triangle: given $(x, y, z_{\text{sp}}) \in T_m$, take

$$u = 1 - x, \quad v = \frac{z_{\text{sp}} - y}{2m}.$$

Using $qx + y + z_{\text{sp}} = 2m$, these formulas recover

$$1 - u = x, \quad 1 + (m - 1)u - mv = y, \quad 1 + (m - 1)u + mv = z_{\text{sp}}.$$

The image of the block ellipsope under $F_{r,\kappa}$ is therefore exactly the set of reduced objective values over T_m . In one direction, lemma 2 sends any $X(u, v) \in \mathcal{E}_m^{\text{blk}}$ to a point of T_m and identifies $F_{r,\kappa}(X(u, v))$ with $\phi_{r,\kappa}(x, y, z_{\text{sp}})$. In the other direction, the inverse coordinates above produce a block matrix in $\mathcal{E}_m^{\text{blk}}$ with the same reduced value.

Likewise, the image of $\mathcal{C}_m^\pm$ under $F_{r,\kappa}$ is exactly the set of reduced objective values over the parity-truncated slice $T_m \cap \{y + z_{\text{sp}} \geq d_m\}$. If $X \in \mathcal{C}_m^\pm$, block exchangeability gives parameters u, v for its second moment, and lemma 6 gives both the reduced-triangle constraints and the parity inequality; lemma 2 again identifies the value with $\phi_{r,\kappa}$. Conversely, any $(x, y, z_{\text{sp}}) \in T_m$ satisfying $d_m \leq y + z_{\text{sp}}$ is converted by the same inverse coordinates into a block matrix $X(u, v)$, and lemma 6 places it in $\mathcal{C}_m^\pm$ with the corresponding reduced objective value.

Substituting these two image descriptions into definition 7 gives

$$\Delta_m^\pm(r, \kappa) = \inf_{\substack{(x,y,z_{\text{sp}}) \in T_m \\ y+z_{\text{sp}} \geq d_m}} \phi_{r,\kappa}(x, y, z_{\text{sp}}) - \inf_{(x,y,z_{\text{sp}}) \in T_m} \phi_{r,\kappa}(x, y, z_{\text{sp}}),$$

as claimed. $\square$

lemma 9 is the bridge from geometry back to design. The relaxed benchmark is the infimum over T_m , while the implementable benchmark is the same reduced objective with the parity truncation imposed. Consequently, the loss is exactly the value difference between these two scalar problems, which is the quantity evaluated by the active-set and truncation formulas above.

B Verification note

This appendix records the verification boundary for the mathematical claims used in the paper. The formal statements take the assignment law, its induced second moments, the two-block weighted graph, the block ellipsope relaxation, and the finite comparison between relaxed and implementable covariance optima as their load-bearing objects. The design-based finite-population potential-outcome framework supplies the surrounding motivation.

The verified mathematical layer consists of the assumptions, definitions, lemmas, and theorems stated in the paper. In particular, it includes the two-block homophily, sign-symmetry, low-scale, and odd-block conditions in assumption 1–assumption 4; the graph, design-class, covariance, objective, loss, and block-sum definitions in definition 1–definition 8; the symmetry reduction in lemma 1; the exactness results in theorem 1 and theorem 2; and the positive-loss and sharp-computation results in theorem 3 and theorem 4. The auxiliary reductions collected in Appendix A are part of the same mathematical layer.

The econometric motivation, related-work positioning, and interpretation are prose surrounding that layer. The displayed assumptions in the paper give exactly the conditions used to derive the formal conclusions, which are finite-dimensional and design based.

For reproducibility, the theorem statements and proofs are machine-checked in Lean 4. The assumed inputs to the checked arguments are the explicitly stated mathematical hypotheses, including the two-block homophily, sign-symmetry, low-scale, odd-block, and nonnegativity conditions where they appear; standard finite-dimensional real matrix primitives for traces, positive semidefiniteness, the Moore–Penrose pseudoinverse, and the Schatten–2 norm; and the usual classical real-analysis and finite-set principles available in the underlying library. The Lean verification discharges the algebraic, convex-geometric, spectral, parity, and active-set claims stated in the formal environments, while the historical, empirical, and interpretive material remains prose.

The formalization was checked against Lean toolchain `leanprover/lean4:v4.29.0-rc3` with `mathlib` at revision `bf8875c7`. All declarations live under the module namespace `CausalSmith.Experimentation.EXP.DesignPm1ExactnessBoundaryV1.Research`, and each carries a `@node` tag matching its manuscript label; every stated result is `sorry-free`. The five main results map to Lean declarations as follows.

Manuscript label	Lean file	Declaration
lemma 1	<code>Helpers/SymmetryReduction.lean</code>	<code>symmetry_reduction</code>
theorem 1	<code>Tcut.lean</code>	<code>cut_corner_exactness</code>
theorem 2	<code>Trobust.lean</code>	<code>robust_corner_exactness</code>
theorem 3	<code>Tgap.lean</code>	<code>gap_window</code>
theorem 4	<code>Tsharp.lean</code>	<code>sharp_rho_star</code>

The auxiliary lemmas of Appendix A are declared in the sibling `Helpers/` files under the same namespace, with declaration names matching their manuscript labels (for example `block_spectral_coordinates`, `weighted_simplex_active_set`, `weighted_simplex_truncation`, and `rounding_gap_reduction`).

C Proofs of the main results

The arguments below are presented as human-readable proof sketches: they give the construction and the load-bearing steps, but the fully formal, gap-free proofs are the Lean 4 declarations listed in Appendix B, which are the authoritative verified artifact. Where a sketch abbreviates a routine algebraic or convex-geometric verification, the complete detail is discharged in the corresponding Lean declaration.

Proof of [lemma 1](#). For the relaxed matrix assertion, let u and v be the averages of the off-diagonal entries of X over, respectively, within-community pairs and across-community pairs, and replace X by the block matrix $X(u, v)$. The unit diagonal is built into $X(u, v)$, and the same two block sums are preserved by the definitions of u and v . To verify membership in the block elliptope, positivity and the unit diagonal first give $|X_{ij}| \leq 1$ for every pair (i, j) , by testing the two vectors $e_i - e_j$ and $e_i + e_j$. Averaging this upper bound over the within-community off-diagonal pairs gives $1 - u \geq 0$. The remaining two inequalities follow by applying positive semidefiniteness of X to the block-sign vector and to the all-ones vector, which give the nonnegativity of $1 + (m - 1)u - mv$ and $1 + (m - 1)u + mv$, respectively. Together with the standing two-block homophily assumption, this proves $X(u, v) \in \mathcal{E}_m^{\text{blk}}$.

The trace terms in $F_{r,\kappa}$ are unchanged by this replacement: L_m , $L_m^\dagger$, and J_n are block-constant, so their traces against a unit-diagonal symmetric matrix depend only on the diagonal

sum and on the within- and across-community off-diagonal sums. The Frobenius norm of $X(u, v)$ is no larger than that of X , by Cauchy–Schwarz applied separately to the two families of off-diagonal entries. Since $\kappa \geq 0$,

$$F_{r,\kappa}(X(u, v)) \leq F_{r,\kappa}(X).$$

For an assignment law $P \in \mathcal{P}_m^{\text{bal}}$, take its orbit average under the full two-block automorphism group. The averaged law remains sign-symmetric and is invariant under permutations within the two communities and under swapping the communities, hence lies in $\mathcal{P}_m^{\text{sym}}$. Sign-symmetry gives zero one-point margins for the averaged law. Its second-moment matrix is the corresponding orbit average of the original second-moment matrix. Because the automorphisms permute the within-community and across-community pair sets, the two block sums of this second-moment matrix agree with those of the original one. Its Frobenius norm is no larger, by the same square-sum averaging argument together with invariance of the total square sum under simultaneous row-column permutations. Therefore the trace terms in $F_{r,\kappa}$ are unchanged, the Frobenius term weakly decreases, and

$$F_{r,\kappa}(X(P^{\text{orb}})) \leq F_{r,\kappa}(X(P)).$$

It remains to pass from pointwise reduction to the two infimum identities. On the relaxed side, the block elliptope is contained in the unit-diagonal positive semidefinite elliptope, it is nonempty because $X(0, 0)$ belongs to it, the objective is bounded below on the relaxed elliptope, and every feasible relaxed matrix has the block-averaged representative above with no larger objective. The elementary infimum-reduction principle therefore gives equality of the relaxed infima. On the implementable side, $\mathcal{P}_m^{\text{sym}} \subseteq \mathcal{P}_m^{\text{bal}}$, the iid design supplies a nonempty symmetric class, assignment second moments are positive semidefinite with unit diagonal and hence inherit the same lower bound, and every balanced design is dominated by its orbit average as above. Applying the same infimum-reduction principle gives the implementable infimum identity. $\square$

Proof of [theorem 1](#). 1. By the two-block homophily assumption, $m \geq 2$, $0 < b < a$. Hence

$$q = 2(m - 1) > 0, \quad a - b > 0, \quad a + b > 0, \quad 2b > 0.$$

Set

$$A = 2b(a + b) \left(1 - \frac{\kappa}{a - b} \right), \quad B = 2b(2m - 2b - \kappa).$$

From $r < \max\{0, \min(A, B)\}$ and $0 \leq r$, we first get

$$r < \min(A, B).$$

Indeed, if $\min(A, B) \leq 0$, then the outer maximum is 0, contradicting $0 \leq r < 0$. Otherwise the outer maximum is $\min(A, B)$. Thus

$$r < A, \quad r < B.$$

2. We now convert these two inequalities into the coefficient separations needed at the cut vertex. From $r < A$, division by the positive number $2b(a + b)$ gives

$$\frac{r}{2b(a + b)} < 1 - \frac{\kappa}{a - b}.$$

Multiplying by $a - b > 0$ and rearranging gives

$$\kappa < (a - b) \left(1 - \frac{r}{2b(a + b)} \right).$$

Using

$$(a + b) + \frac{r}{a + b} - \left(2b + \frac{r}{2b} + \kappa \right) = (a - b) \left(1 - \frac{r}{2b(a + b)} \right) - \kappa$$

and $c_x/q = (a + b) + r/(a + b)$, $c_y = 2b + r/(2b)$, we obtain

$$\frac{c_x}{q} > c_y + \kappa.$$

Similarly, from $r < B$, division by $2b > 0$ gives

$$\frac{r}{2b} < 2m - 2b - \kappa,$$

and hence

$$2b + \frac{r}{2b} + \kappa < 2m.$$

Since $c_z = 2m$, this is

$$c_z > c_y + \kappa.$$

3. The hypotheses of [lemma 3](#) are therefore satisfied: $q > 0$, $\kappa \geq 0$, and the two coefficient separations above hold. Hence

$$(0, 2m, 0) \in T_m,$$

and every other point of T_m has strictly larger reduced objective value.

By [lemma 2](#), the cut covariance is

$$X_{\text{cut}} = X(1, -1),$$

and its spectral coordinates are

$$(1 - 1, 1 + (m - 1) \cdot 1 - m(-1), 1 + (m - 1) \cdot 1 + m(-1)) = (0, 2m, 0).$$

Thus $X_{\text{cut}} \in \mathcal{E}_m^{\text{blk}}$, and

$$F_{r,\kappa}(X_{\text{cut}}) = \phi_{r,\kappa}(0, 2m, 0).$$

4. Let $X \in \mathcal{E}_m^{\text{blk}}$ and suppose $X \neq X_{\text{cut}}$. Write $X = X(u, v)$. By [lemma 2](#), the associated coordinates

$$x = 1 - u, \quad y = 1 + (m - 1)u - mv, \quad z_{\text{sp}} = 1 + (m - 1)u + mv$$

belong to T_m , and

$$F_{r,\kappa}(X) = \phi_{r,\kappa}(x, y, z_{\text{sp}}).$$

These coordinates cannot equal $(0, 2m, 0)$: if $x = 0$, then $u = 1$, and if also $z_{\text{sp}} = 0$, then

$$m(1 + v) = 0,$$

so $v = -1$ since $m > 0$, giving $X = X_{\text{cut}}$, a contradiction. The strict part of [lemma 3](#) therefore gives

$$F_{r,\kappa}(X_{\text{cut}}) = \phi_{r,\kappa}(0, 2m, 0) < \phi_{r,\kappa}(x, y, z_{\text{sp}}) = F_{r,\kappa}(X).$$

Thus X_{cut} is the unique minimizer over $\mathcal{E}_m^{\text{blk}}$.

5. The cut design is the two-point cut law, so

$$P_{\text{cut}} \in \mathcal{P}_m^{\text{sym}}, \quad X(P_{\text{cut}}) = X_{\text{cut}}.$$

Consequently $X_{\text{cut}} \in \mathcal{C}_m^\pm$. Moreover, every $X \in \mathcal{C}_m^\pm$ is obtained from a block-exchangeable design. Its second moment is block-symmetric, say $X = X(u, v)$, and the forward reduced-slice characterization places its spectral coordinates in the same reduced triangle T_m . By [lemma 2](#), this implies

$$\mathcal{C}_m^\pm \subseteq \mathcal{E}_m^{\text{blk}}.$$

Therefore, if $X \in \mathcal{C}_m^\pm$ and $X \neq X_{\text{cut}}$, the strict relaxed optimality already proved yields

$$F_{r,\kappa}(X_{\text{cut}}) < F_{r,\kappa}(X).$$

Thus X_{cut} is also the unique minimizer over $\mathcal{C}_m^\pm$.

6. It remains to identify the gap. From the strict inequalities above, with equality allowed only at X_{cut} , we have the non-strict minimizing inequalities

$$F_{r,\kappa}(X_{\text{cut}}) \leq F_{r,\kappa}(X) \quad \text{for all } X \in \mathcal{E}_m^{\text{blk}},$$

and

$$F_{r,\kappa}(X_{\text{cut}}) \leq F_{r,\kappa}(X) \quad \text{for all } X \in \mathcal{C}_m^\pm.$$

Since X_{cut} belongs to both feasible classes, the infimum of each corresponding objective image is exactly $F_{r,\kappa}(X_{\text{cut}})$. Hence, by the definition of the implementability gap in [definition 7](#),

$$\Delta_m^\pm(r, \kappa) = F_{r,\kappa}(X_{\text{cut}}) - F_{r,\kappa}(X_{\text{cut}}) = 0.$$

This proves all asserted conclusions. □

Proof of [theorem 2](#). 1. We first prove the finite-weight equivalence. Write

$$q = 2(m-1), \quad c_x = q \left((a+b) + \frac{r}{a+b} \right), \quad c_y = 2b + \frac{r}{2b}, \quad c_z = 2m.$$

Suppose that some $\kappa \geq 0$ makes I_n a relaxed minimizer over $\mathcal{E}_m^{\text{blk}}$. For an arbitrary $(x, y, z_{\text{sp}}) \in T_m$, set

$$u = 1 - x, \quad v = \frac{z_{\text{sp}} - y}{2m}.$$

Using [lemma 2](#), this inverse change of variables sends (x, y, z_{sp}) to a feasible block matrix $X(u, v)$, and the matrix objective equals the reduced objective at (x, y, z_{sp}) . Since $I_n = X(0, 0)$ corresponds to $(1, 1, 1)$, the assumed matrix minimality says that $(1, 1, 1)$ minimizes

$$c_x x + c_y y + c_z z_{\text{sp}} + \kappa \sqrt{q x^2 + y^2 + z_{\text{sp}}^2}$$

on T_m . The certificate-necessity part of [lemma 5](#) therefore gives

$$\frac{c_x}{q} = c_y, \quad c_y = c_z.$$

The first equality is

$$a + b + \frac{r}{a + b} = 2b + \frac{r}{2b},$$

hence

$$(a - b) \left(1 - \frac{r}{2b(a + b)} \right) = 0.$$

By two-block homophily, $a > b > 0$, so

$$r = 2b(a + b).$$

Substituting this into $c_y = c_z$ gives

$$2b + \frac{2b(a + b)}{2b} = 2m,$$

that is $a + 3b = 2m$. Thus the affine-balanced locus holds. Conversely, if

$$a + 3b = 2m, \quad r = 2b(a + b),$$

then the same algebra gives $c_x/q = c_y = c_z$. Taking $\kappa = 1$, the strict optimality part of [lemma 5](#), transported through [lemma 2](#), gives $I_n \in \mathcal{E}_m^{\text{blk}}$ and strict optimality against every $X \neq I_n$. For $X = I_n$ the desired inequality is equality, while for $X \neq I_n$ it follows from strict optimality. Hence a finite nonnegative robustness weight exists exactly on the stated locus.

2. On the affine-balanced locus, let $\kappa > 0$. As above,

$$\frac{c_x}{q} = c_y = c_z.$$

The point $I_n = X(0, 0)$ has reduced coordinates $(1, 1, 1)$ by [lemma 2](#). The Frobenius-center certificate, [lemma 5](#), gives strict reduced optimality of $(1, 1, 1)$ on T_m . Translating back by [lemma 2](#), $I_n \in \mathcal{E}_m^{\text{blk}}$, and for any $X = X(u, v) \in \mathcal{E}_m^{\text{blk}}$ with $X \neq I_n$, the associated reduced coordinates are not $(1, 1, 1)$; otherwise $u = 0$ and $v = 0$, hence $X = I_n$. Therefore

$$F_{r,\kappa}(I_n) < F_{r,\kappa}(X).$$

3. The iid law is invariant under sign flip, within-block permutations, and block swap, so

$$P_{\text{iid}} \in \mathcal{P}_m^{\text{sym}}.$$

Its second moments are diagonal: the diagonal entries equal 1, and for $i \neq j$ the involution that flips the i -th sign pairs every assignment with one contributing the negative product. Thus all off-diagonal moments vanish and

$$X(P_{\text{iid}}) = I_n.$$

4. Now assume the affine-balanced locus fails. If for some $\kappa > 0$ the matrix I_n were a relaxed minimizer, then the same center-coefficient necessity argument from the first step would give

$$\frac{c_x}{q} = c_y, \quad c_y = c_z,$$

and the same algebra would force

$$a + 3b = 2m, \quad r = 2b(a + b),$$

contradicting the assumption. Hence I_n is not a relaxed minimizer at any positive finite robustness weight off the locus.

5. Finally, let $(X_\kappa)_{\kappa \in \mathbb{R}}$ be any family of relaxed minimizers for positive κ . Since $r \geq 0$ and $a > b > 0$, the reduced linear coefficients c_x, c_y, c_z are nonnegative. Comparing each minimizer with the feasible point I_n gives

$$F_{r,\kappa}(X_\kappa) \leq F_{r,\kappa}(I_n).$$

Writing $X_\kappa = X(u_\kappa, v_\kappa)$, the reduced squared norm decomposes as

$$q(1 - u_\kappa)^2 + (1 + (m - 1)u_\kappa - mv_\kappa)^2 + (1 + (m - 1)u_\kappa + mv_\kappa)^2 = q + 2 + N_{\text{same}}u_\kappa^2 + N_{\text{cross}}v_\kappa^2,$$

with positive constants N_{same} and N_{cross} . If a same-block off-diagonal entry stayed at distance at least $\varepsilon > 0$ from zero for arbitrarily large κ , then $|u_\kappa| \geq \varepsilon$, so the robustness norm would exceed its value at I_n by a fixed amount $\delta > 0$. The nonnegative linear part then yields

$$\kappa(\|I_n\| + \delta) \leq F_{r,\kappa}(X_\kappa) \leq F_{r,\kappa}(I_n) = L + \kappa\|I_n\|,$$

forcing $\kappa\delta \leq L$, impossible for all sufficiently large κ . The same argument with N_{cross} shows that the across-block parameter $v_\kappa \rightarrow 0$. Diagonal entries are identically 1, while off-diagonal entries are either u_κ or v_κ . Hence every entry of X_κ converges to the corresponding entry of I_n as $\kappa \rightarrow \infty$. □

Proof of [theorem 3](#). 1. [Assumption 1](#) gives $m \geq 2$, $b < a$, and $b > 0$. Hence

$$q = 2(m - 1) > 0, \quad \sqrt{q} > 0, \quad a + b > 0, \quad a - b > 0.$$

[assumption 3](#) gives

$$2m - a - 3b > 0.$$

These inequalities are the standing sign facts used below for division and multiplication by positive quantities.

2. Assume now that [assumption 4](#) holds, and fix

$$0 \leq \kappa < \kappa_{\text{gap}}(m, a, b).$$

Unfolding the definition of κ_{gap} and multiplying by the positive factor $(a + b)/((a - b)\sqrt{q})$, we get

$$\frac{\kappa(a + b)}{(a - b)\sqrt{q}} < 2m - a - 3b.$$

Therefore

$$R_x^+ - R_x^- = (a + b) \left((2m - a - 3b) - \frac{\kappa(a + b)}{(a - b)\sqrt{q}} \right) > 0.$$

Thus $R_x^- < R_x^+ = r_{\text{gap}}^+$, and since

$$r_{\text{gap}}^- = \frac{R_x^- + R_x^+}{2},$$

we have

$$R_x^- < r_{\text{gap}}^- < r_{\text{gap}}^+.$$

Also $R_x^- \geq 0$, and

$$2b(a+b) \left(1 - \frac{\kappa}{a-b}\right) \leq R_x^-,$$

so

$$r_{\text{cut,gap}}(m, a, b, \kappa) = \max \left\{ 0, 2b(a+b) \left(1 - \frac{\kappa}{a-b}\right) \right\} \leq R_x^-.$$

Combining these inequalities yields

$$r_{\text{cut,gap}}(m, a, b, \kappa) < r_{\text{gap}}^-(m, a, b, \kappa) < r_{\text{gap}}^+(m, a, b, \kappa).$$

3. Let $r_{\text{gap}}^- < r < r_{\text{gap}}^+$. From the preceding step,

$$R_x^- < r.$$

After division by $2b(a+b) > 0$, this says

$$1 + \frac{\kappa}{(a-b)\sqrt{q}} < \frac{r}{2b(a+b)}.$$

Multiplying by $a-b > 0$ gives

$$\frac{\kappa}{\sqrt{q}} < (a-b) \left(\frac{r}{2b(a+b)} - 1 \right).$$

Using the identity

$$2b + \frac{r}{2b} - \left(a + b + \frac{r}{a+b} \right) = (a-b) \left(\frac{r}{2b(a+b)} - 1 \right),$$

and $c_x/q = a + b + r/(a+b)$, this proves

$$c_y > \frac{c_x}{q} + \frac{\kappa}{\sqrt{q}}.$$

Similarly, the upper bound $r < r_{\text{gap}}^+ = R_x^+$ gives

$$\frac{r}{a+b} < 2m - a - b - \frac{\kappa}{\sqrt{q}},$$

hence, since $c_z = 2m$ and $c_x/q = a + b + r/(a+b)$,

$$c_z > \frac{c_x}{q} + \frac{\kappa}{\sqrt{q}}.$$

By the spread-vertex certificate, [lemma 4](#), the point

$$\left(\frac{2m}{q}, 0, 0 \right) = \left(\frac{m}{m-1}, 0, 0 \right)$$

belongs to T_m and is the strict unique minimizer of the reduced objective on T_m .

4. The spread covariance is $X(-1/(m-1), 0)$, whose spectral coordinates are

$$x = 1 - \left(-\frac{1}{m-1}\right) = \frac{2m}{q}, \quad y = 0, \quad z_{\text{sp}} = 0.$$

By [lemma 2](#), the membership of this triple in T_m gives

$$X_{\text{spread}} \in \mathcal{E}_m^{\text{blk}},$$

and the value of $F_{r,\kappa}$ at X_{spread} is the reduced objective at $(2m/q, 0, 0)$. If $X \in \mathcal{E}_m^{\text{blk}}$, write $X = X(u, v)$. [Lemma 2](#) sends it to a point of T_m and identifies $F_{r,\kappa}(X)$ with the reduced objective there. If this reduced point equals $(2m/q, 0, 0)$, then $y = z_{\text{sp}} = 0$ forces $v = 0$ and $u = -1/(m-1)$, hence $X = X_{\text{spread}}$. Therefore every $X \neq X_{\text{spread}}$ corresponds to a different point of T_m , and the strict reduced optimality from Step 3 gives

$$F_{r,\kappa}(X_{\text{spread}}) < F_{r,\kappa}(X).$$

5. Since m is odd, [lemma 6](#) imposes the implementability condition

$$y + z_{\text{sp}} \geq \frac{2}{m}.$$

The spread point has $y + z_{\text{sp}} = 0$, so

$$X_{\text{spread}} \notin \mathcal{C}_m^{\pm}.$$

For the loss, pass from (x, y, z_{sp}) to simplex variables

$$t = (qx, y, z_{\text{sp}}), \quad t_x + t_y + t_z = 2m,$$

with weights $\beta = (1/q, 1, 1)$ and coefficients

$$\alpha = \left(\frac{c_x}{q}, c_y, c_z\right).$$

The relaxed minimizer is $t^{\text{rel}} = (2m, 0, 0)$, and it violates the odd parity constraint $t_y + t_z \geq 2/m$. [Lemma 8](#) gives a minimizer on the truncated simplex, hence an implementable reduced minimizer. Since the relaxed minimizer is unique and lies outside the truncated slice, that implementable minimizer has strictly larger reduced objective value. Finally [lemma 9](#) identifies

$$\Delta_m^{\pm}(r, \kappa)$$

with the truncated reduced infimum minus the relaxed reduced infimum, so

$$\Delta_m^{\pm}(r, \kappa) > 0.$$

6. It remains to prove the even case. Assume m is even and let $X \in \mathcal{E}_m^{\text{blk}}$. Write $X = X(u, v)$, and let (x, y, z_{sp}) be its spectral coordinates. By [lemma 2](#), this point lies in T_m , so $y, z_{\text{sp}} \geq 0$. For even m , the parity threshold in [lemma 6](#) is 0, and hence $y + z_{\text{sp}} \geq 0$ automatically. The same lemma therefore gives $X \in \mathcal{C}_m^{\pm}$. Thus

$$\mathcal{E}_m^{\text{blk}} \subseteq \mathcal{C}_m^{\pm}.$$

□

Proof of [theorem 4](#). **1. Equality and range.** By definition, $\rho_\star(m, a, b, r, \kappa)$ is the difference between the implementable reduced value and the relaxed reduced value. The reduced gap identity in [lemma 9](#) identifies this same difference with $\Delta_m^\pm(r, \kappa)$. Hence

$$\rho_\star(m, a, b, r, \kappa) = \Delta_m^\pm(r, \kappa).$$

The nonnegativity of $\rho_\star$ is the constrained-infimum comparison for the reduced problem: the implementable feasible set is the parity-truncated subset of T_m , while the relaxed feasible set is all of T_m . The objective is bounded below using $r \geq 0$, $\kappa \geq 0$, the nonnegative reduced coordinates, and $a > b > 0$. Applying the same reduced identity gives the nonnegativity of $\Delta_m^\pm(r, \kappa)$.

2. Zero loss. Again using the definition of $\rho_\star$ together with [lemma 9](#), zero rounding loss is exactly zero reduced implementability gap. The zero-gap criterion says that this happens precisely when some relaxed minimizer over T_m also belongs to the parity-truncated slice, namely when there is $(x, y, z_{\text{sp}}) \in T_m$ minimizing $\phi_{r, \kappa}$ over all of T_m and satisfying

$$d_m \leq y + z_{\text{sp}}.$$

This is the stated zero-loss equivalence.

3. Strict robustness. Assume $\kappa > 0$. Since $m \geq 2$ under [assumption 1](#), we have $q = 2(m - 1) > 0$. Under the change of variables

$$t = (qx, y, z_{\text{sp}}), \quad \alpha^\# = \left(\frac{c_x}{q}, c_y, c_z\right), \quad \beta^\# = \left(\frac{1}{q}, 1, 1\right),$$

the reduced triangle T_m is the simplex of total mass $2m$, and $\phi_{r, \kappa}$ becomes the weighted-simplex objective. The positive-robustness part of [lemma 7](#) gives a unique weighted-simplex minimizer, hence a unique relaxed minimizer of $\phi_{r, \kappa}$ on T_m . With uniqueness in hand, the zero-gap criterion from Step 2 is equivalent to saying that every relaxed minimizer satisfies $d_m \leq y + z_{\text{sp}}$.

4. Linear case. Assume $\kappa = 0$, and fix $(x, y, z_{\text{sp}}) \in T_m$. Passing to $t = (qx, y, z_{\text{sp}})$, the reduced objective is the linear weighted-simplex objective with coefficients $\alpha^\#$. The zero-robustness part of [lemma 7](#) identifies the minimizers with the exposed face supported only on coordinates whose $\alpha^\#$ -coefficient is minimal. Translating this exposed-face condition back to (x, y, z_{sp}) gives exactly the three displayed implications: positive x -mass requires $c_x/q \leq c_y, c_z$, positive y -mass requires $c_y \leq c_x/q, c_z$, and positive z_{sp} -mass requires $c_z \leq c_x/q, c_y$.

5. Active-set formula. Assume $\kappa > 0$. The same active-set construction from Step 3 supplies an admissible support $S \subseteq \{x, y, z_{\text{sp}}\}$ and multiplier λ for $\alpha^\#, \beta^\#, \kappa$. The corresponding simplex point $t^{S, \lambda}$ has mass $2m$; therefore

$$\left(\frac{t_x^{S, \lambda}}{q}, t_y^{S, \lambda}, t_z^{S, \lambda}\right) \in T_m.$$

The weighted-simplex strict minimizer property transfers back through $t = (qx, y, z_{\text{sp}})$, so this reduced point minimizes $\phi_{r, \kappa}$ over T_m . The value formula in [lemma 7](#) gives the relaxed reduced value $2m\lambda$.

6. Truncation correction. Suppose no relaxed minimizer in T_m satisfies $d_m \leq y + z_{\text{sp}}$. Choose a relaxed minimizer and pass to $t = (qx, y, z_{\text{sp}})$. Its weighted-simplex coordinates minimize the unconstrained simplex problem but fail the truncated condition $t_y + t_z \geq d_m$. [Lemma 8](#) then puts the constrained minimizer on the boundary segment

$$H_{d_m} = \{(2m - d_m, s, d_m - s) : 0 \leq s \leq d_m\}$$

at

$$s_\star = \text{truncSelector}(2m, d_m, \alpha^\sharp, \beta^\sharp, \kappa).$$

Converting the resulting constrained weighted-simplex optimum back to the reduced notation gives

$$\text{val}_{\text{impl}}(m, a, b, r, \kappa) = \Phi_{\alpha^\sharp, \beta^\sharp, \kappa}(2m - d_m, s_\star, d_m - s_\star).$$

7. Even block size. If m is even, then $d_m = 0$. A relaxed minimizer exists, and every point of T_m has $y, z_{\text{sp}} \geq 0$, hence $0 = d_m \leq y + z_{\text{sp}}$. The zero-loss criterion from Step 2 therefore gives

$$\rho_\star(m, a, b, r, \kappa) = 0.$$

□

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